User's Guide for the COHIWizard

Version V2.3.7 and higher for Windows 10/11 and LINUX, 2026/11/12

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1 DOCUMENTATION

1.1 SUPPORTED OPERATING SYSTEMS AND INSTALLATION OPTIONS

The COHIWizard is available for Windows 10/11 (64 bit) and LINUX (tested with Debian 12 and 13 on x86_64 PCs). From version 2.3 on it is always executed from its Python sources in a virtual Python environment of its own. Installation, updates and the selection of modules and device drivers are done by the setup tool cohi_setup.py, which downloads the required files directly from the GitHub repository. It can be started in three ways:

- **Windows:** with the installer COHIWizard_Setup_2.3.exe (recommended). It brings its own Python and needs no administrator rights.
- **Windows, alternative:** with cohi_setup.bat, using an installed Python.
- **LINUX:** with cohi_setup.py and the system Python.

The zip file with the executable SDR_COHIWizard_v2.x.exe used for the versions up to 2.2 is no longer needed.

1.2 INSTALLATION

1.2.1 Windows: installer

Download the installer COHIWizard_Setup_2.3.exe from the COHIRADIA software page and start it. The installer needs no administrator rights and installs for the current user:

- the installation directory, by default %USERPROFILE%\COHIWizard. It is called 'COHI_root' in this documentation;
- a private Python 3.10 in COHI_root\python (nothing is registered in Windows);
- the current version of the setup tool cohi_setup.py, which it then starts.

A console window opens in which the setup tool creates the virtual environment COHI_root\venv and installs the required Python packages. **Depending on your computer this can take a while, on some older PCs even 10 – 30 minutes,** please be patient. Then the window 'COHIWizard setup' opens (Fig. *1-1*), in which you select the modules and drivers and press 'Install / apply changes' (see below).

After the installation the start menu contains the entries 'COHIWizard', 'COHIWizard (with console for error messages)', 'COHIWizard Setup (manage modules and drivers)' and the uninstaller. The uninstaller removes Python, the virtual environment and the launchers and asks whether COHI_root\sources (including config_wizard.yaml and all data stored there) should be deleted as well.

Alternative Python version:

Alternatively, if you prefer a pure Python installation without an executable, but COHIWizard.py instead, then sources\cohi_setup.bat can be used. It looks for an installed Python 3.10 – 3.13 (if none is found, it installs Python 3.10.11 for the current user), fetches the current cohi_setup.py from GitHub and starts it.

1.2.2 LINUX

Download the setup tool cohi_setup.py (https://raw.githubusercontent.com/hermy-sf/COHIWizard/cohiwizard_v2.3/sources/cohi_setup.py) and start it in a terminal with the system Python (3.10 – 3.13):

```
python3 cohi_setup.py
```

The tool asks for the installation directory (default ~/COHIWizard), creates the virtual environment, installs the Python packages and opens the window 'COHIWizard setup'. If the Python module venv is missing (Debian/Ubuntu: package python3-venv), it offers to install it. Further system packages are installed with apt after a query as well: ffmpeg, the osmo-fl2k packages and a udev rule for the FL2K dongle, and a udev rule for the ADALM2000. Root rights are requested via sudo or pkexec. If you decline a query, the necessary commands are listed at the end.

The setup tool creates the launchers COHIWizard.sh and COHIWizard_setup.sh in COHI_root and a menu entry (~/.local/share/applications/cohiwizard.desktop). Without a graphical desktop it can also be used from the command line, e.g. python3 cohi_setup.py --no-gui --modules synthesizer,annotator --drivers fl2k_universal (python3 cohi_setup.py --help lists all options).

1.2.3 The setup window: modules and drivers

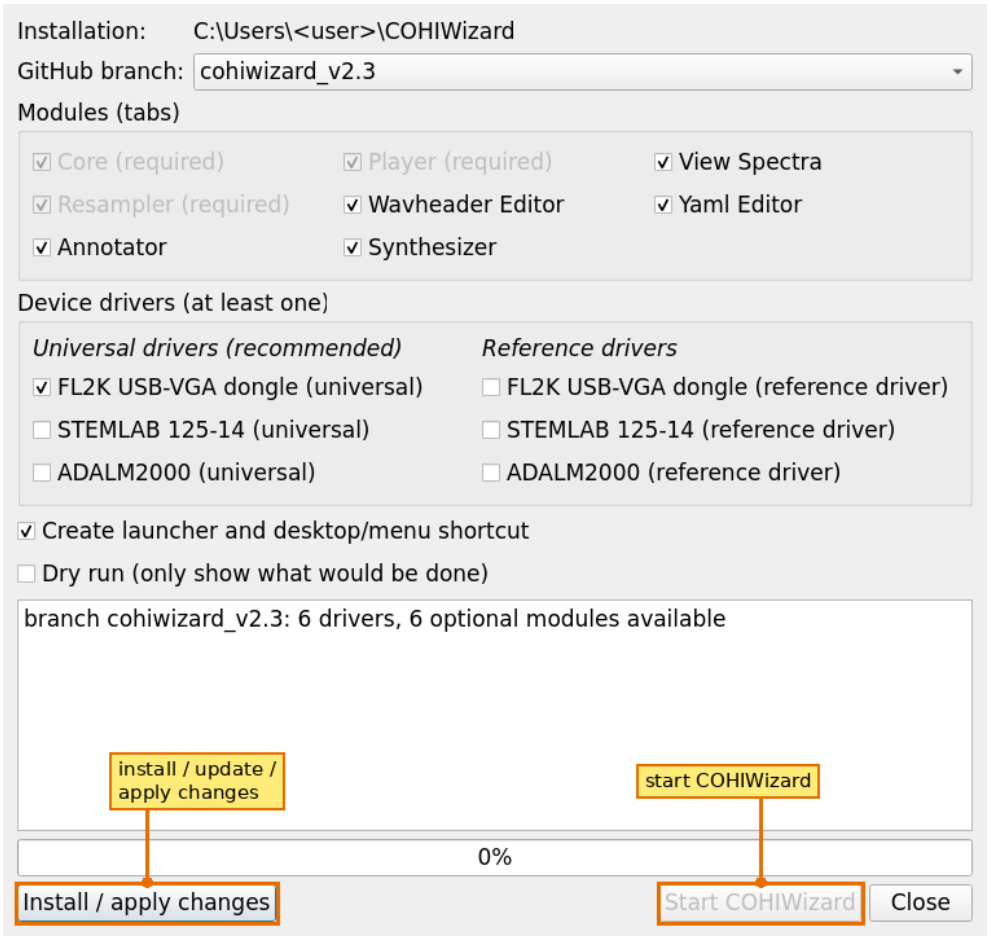


Fig. 1-1: The window of the setup tool

In the setup window you select:

- **GitHub branch:** main contains the stable release. Development branches (e.g. cohiwizard_v2.3) contain the newest functions but are less tested.
- **Modules (tabs):** Core, Player and Resampler are always installed, the other modules are optional. Deselected modules are removed and their tab disappears; they can be installed again at any time.

- **Device drivers:** at least one driver must be selected. The universal drivers are recommended; the reference drivers are the drivers of v2.2 (see section 1.5.5). Drivers that are deselected are removed.
- **Create launcher and desktop/menu shortcut** and **Dry run** (only shows what would be done). Remark: checking 'dry run' does not install anything, so this is only a testing mode.

'Install / apply changes' downloads the selected files from GitHub (git is not needed) and shows the progress in the log window. Only new or changed files are downloaded, therefore the same button is also used for updates. The setup tool also installs the external software required by the selected drivers: ffmpeg (under Windows as a local copy in sources\ffmpeg-master-latest-win64-gpl-shared), the osmo-fl2k library for the FL2K drivers, libm2k for the ADALM2000 drivers and, under Windows, the USB driver tool Zadig for the FL2K dongle (see section 1.5.5). Every installation of external software is confirmed by a query. Finally 'Start COHIWizard' starts the program.

1.2.4 Updating and changing the installation

Start the setup tool again to update the COHIWizard or to add or remove modules and drivers: via the start menu entry 'COHIWizard Setup', COHIWizard_setup.bat / COHIWizard_setup.sh in COHI_root, or from the running COHIWizard with the menu 'Setup → Install / remove modules and drivers ...' (Fig. *1-2*). The COHIWizard is closed first and can be started again from the setup window afterwards. Your settings in config_wizard.yaml are kept.

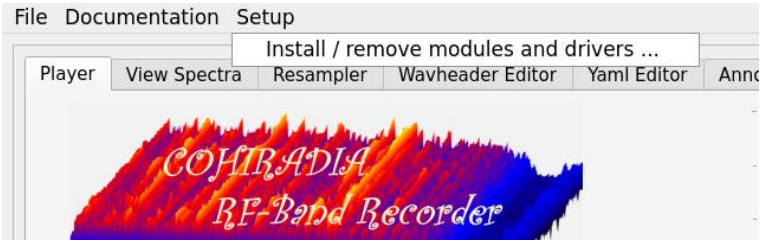


Fig. 1-2: Starting the setup tool from the COHIWizard

Developers: If you prefer to work with a clone of the GitHub repository, follow the instructions in the README file. The setup tool does not change a git checkout; it only performs a dry run there.

1.3 Available Tabs

There are several tabs on the top of the GUI which allow to select the different tools of the COHIWizard. Mandatory are the tabs 'Player' and 'Resampler' (the latter is also needed internally). Optional are 'View Spectra', 'Wavheader Editor', 'Yaml Editor', 'Annotator' and 'Synthesizer'. Which tabs are present depends on the modules selected in the setup tool (section 1.2). Some users may wish to have only the most important tabs visible, or some tabs may be hardly usable on the computer due to resource limitations, e.g. on a Raspberry Pi. Simply deselect these modules in the setup tool; they can be installed again at any time.

The setup tool writes the configuration file 'config_modules.yaml' automatically. Editing it manually (section 3.2) is only necessary in special cases. WARNING: Before changing config_modules.yaml manually, make a backup of the file. If configured wrongly, the Wizard may not start any more correctly; running the setup tool again restores a valid file.

1.4 STARTING THE APP

Before you begin: The COHIWizard app can have different appearances ('skins'), depending on the computer platform it is being executed on. Since v2.3 there are three skins: **skin 5** is the default for PC screens (see Fig. *1-3*) and scales when the window is resized; skins 2 and 3 are meant for 7" displays or smaller, e.g. on a Raspberry Pi (see Fig. *1-4*). All of them offer the modulator types, the Mixer slider and the audio station lists. The former skins 0 and 1 were removed in v2.3.

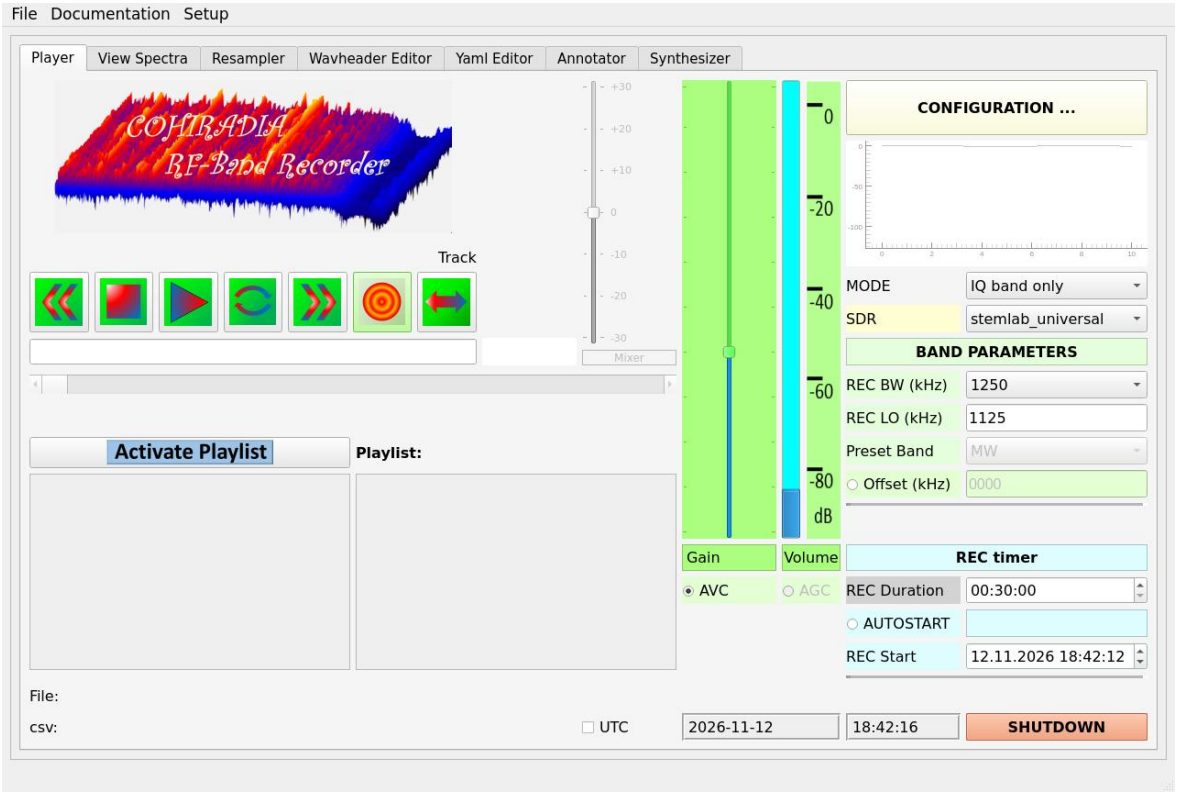


Fig. 1-3: Start view of the Player tab in appearance 'skin 5'

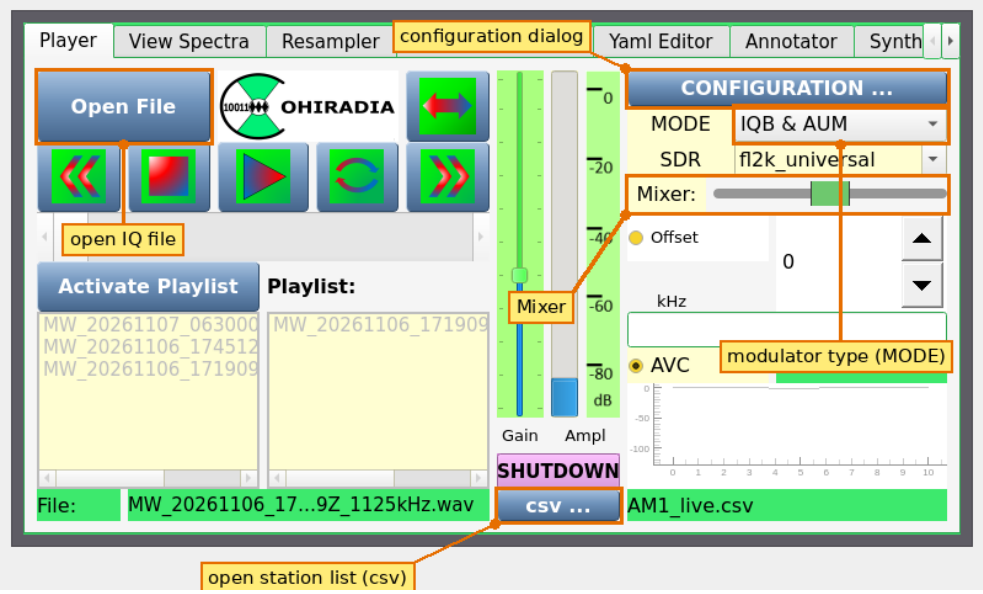


Fig. 1-4: Player tab under skin 2 (IQ band and audio modulator)

There is an automatism which starts skin 2 automatically when the app is run with a display with 7" or smaller. If the display is larger, then the skin used depends on the value of the settings variable 'skinindex' in the file config_wizard.yaml (default 5, see section 3.3). If the automatic choice of skin 2 on 7" displays is unwanted, then the user can change the variable 'autoskin' to 'False'. Most figures in this user's guide show skin 5. Skins 2 and 3 have a reduced set of control elements with larger buttons for touchscreens; functions which are unlikely to be used on a Raspberry Pi (e.g. recording) are left out.

Windows: Start the app with the start menu entry 'COHIWizard', the desktop shortcut or COHIWizard.bat in COHI_root. The program runs without a console window; its messages are written to COHI_root\sources\COHIWizard_console.log. If the program does not behave as expected, start it with 'COHIWizard (with console for error messages)' or COHIWizard_console.bat: the console window stays open and shows the error messages.

LINUX: Start the app with the menu entry 'COHIWizard' or with COHI_root/COHIWizard.sh.

If you prefer a manual installation: open a terminal, activate the virtual environment (source COHI_root/venv/bin/activate), change to COHI_root/sources and type python COHIWizard.py.

The program window opens on the 'Player' tab (or on the tab selected as 'Startup Tab' in the configuration dialog) and shows the control and display elements of the player (Fig. *1-3*).

The user manual can be opened directly from the GUI, clicking onto the menu 'Documentation' and selecting 'open pdf', see Fig. *1-5*. ATTENTION: This option is not available in skin 2 and 3 !

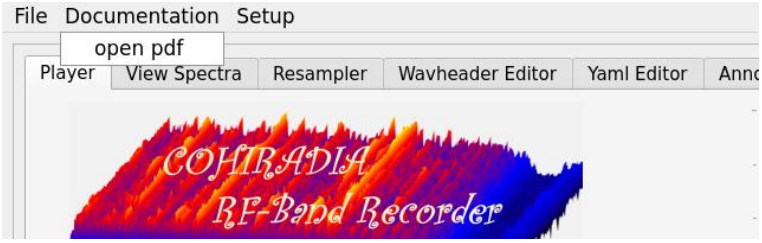


Fig. 1-5: Opening the user manual

Before playing back a recording you have to open it by clicking on the 'File' menu in the left upper corner and selecting 'File open', see Fig. *1-6*. Alternatively the shortcut ‚Alt-F' can be used. The same menu contains 'Playlist open' (Alt-L) for loading an audio station list (section 1.5.2). For users which have used v2.2 before: The function 'Overwrite header' is no longer in this menu; it is now a button in the tab 'WAV Header'. In the mode 'audio modulator only' no IQ file is needed.

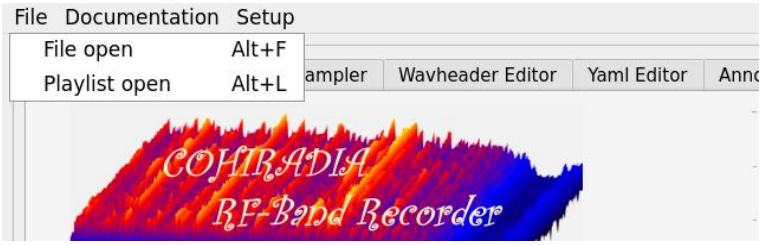


Fig. 1-6: The File menu

When using skin 2 or 3, there is no menu bar on top. Instead, there is a big button 'Open File' on the top left for the IQ file and a button 'csv ...' in the bottom line for the audio station list (see Fig. *1-4*).

If you open another IQ file while the player is running (or paused), the current playback is stopped automatically, the new file is loaded and playback is restarted with it; no confirmation is needed. If you cancel the file selection, the current file simply continues to play. While a recording is running no file can be opened; stop the recording first.

1.5 TAB PLAYER

This tab plays back IQ wav files as well as the old '*.dat' files from the early days of COHIRADIA as an RF signal via the connected SDR hardware. Since v2.3 the player can additionally AM-modulate audio sources (local audio files, m3u playlists, web radio streams) onto freely selectable carrier frequencies, either alone or mixed into the band of an IQ recording (section 1.5.2). With a STEMLAB you can also record your own RF signals to IQ wav files (section 1.5.3).

1.5.1 Player

The Player has the following important features (see Fig. *1-7*):

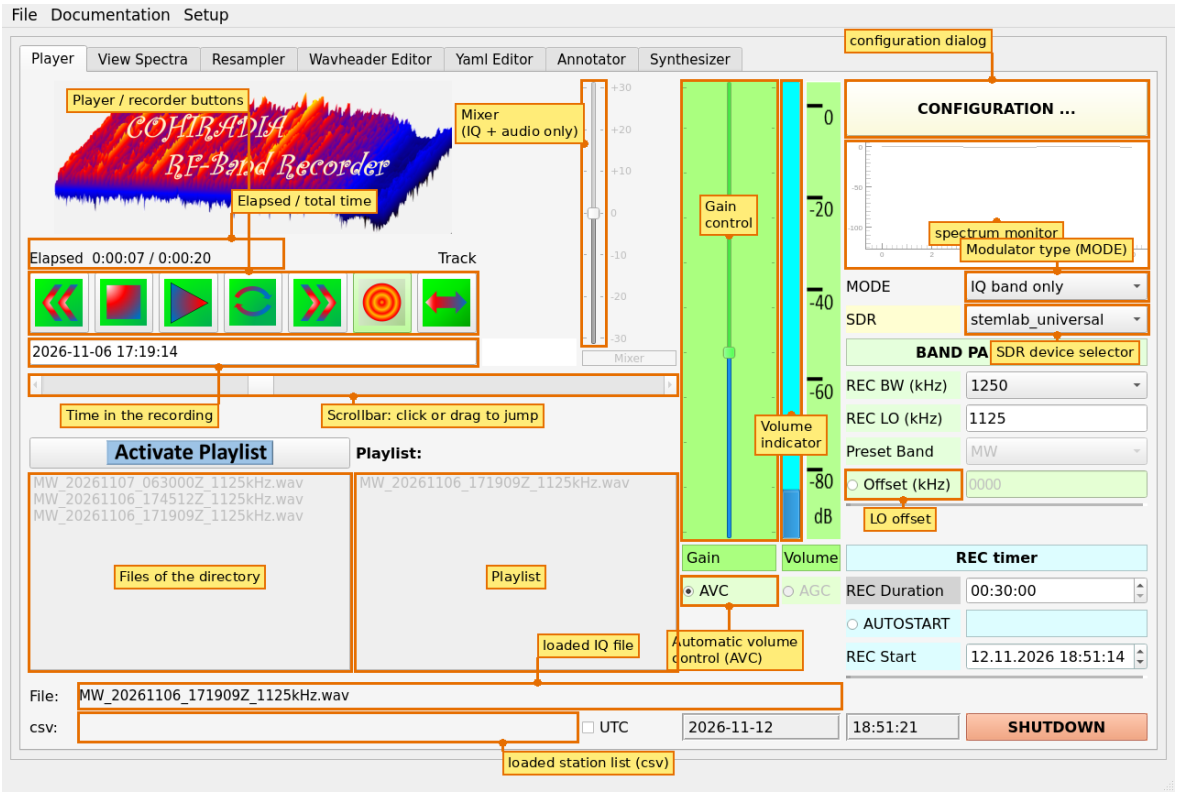


Fig. 1-7: Player tab (skin 5) with an IQ file loaded

(1) Select the modulator type in the combo box 'MODE' and then the playback device in the combo box 'SDR'. The list only contains the installed drivers which support the chosen modulator type (section 1.5.2 and 1.5.5). The last selection is remembered for the next start; directly after the installation 'IQ band only' is preset.

(2) After loading a file, you can use the scroll bar to jump back and forth in the file, making it very easy to locate different points in the recording. You can drag the slider or simply click at the

desired position in the bar. The length of the bar corresponds to the true size of the file, so this also works for files larger than 4 GB and for '*.dat' files.

(3) The current time is displayed in the field above the scroll bar, corresponding to the time fields of the wav header. This only works with wav files, not with '*.dat' files.

(4) Above the transport buttons the line 'Elapsed' (skin 5) shows the time elapsed since the beginning of the file and the total playing time of the file, e.g. 'Elapsed 0:00:07 / 0:00:20'. During a recording only the elapsed time is shown.

(5) There is a logarithmic signal strength indicator ('Volume'), which should optimally display between 70 and 80%. If the signal is too weak (bar turns yellow), you can use the 'Gain' control to increase the gain within limits (logarithmically). If you turn up too much, clipping and intermodulation distortion will occur during playback and the bar will turn red.

(6) The radio button 'AVC' activates the automatic volume control. Once activated (default) the signal strength of the output is kept close to the maximum level at which no intermodulation or clipping occurs. If you wish to set the signal strength manually, deactivate the radio button. For the USB devices (fl2k, ADALM2000) the AVC is strongly recommended. However, in some cases the AVC fails due to an uncommon signal structure. Then AVC should be inactivated and the volume has to be controlled manually.

(7) You can define a playlist for IQ-wav-files and thus automatically play several recordings of a directory one after the other (see below). The directory is selected automatically when opening a file, it is the one in which the file resides.

(8) The transport buttons (Fig. *1-8*) are Rewind, Stop, Play/Pause, Endless play, Fast forward, Record and Manual tracking. They can also be operated with the keyboard (Play: Alt+P, Stop: Alt+X, Record: Alt+R, Rewind: Alt+<, Tracking: Alt+T).

(9) The lines 'File:' and 'csv:' at the bottom show the loaded IQ file and the loaded audio station list. Both remain loaded after STOP, so that you can start again with Play. Pressing STOP once more while the player is stopped unloads the IQ file.

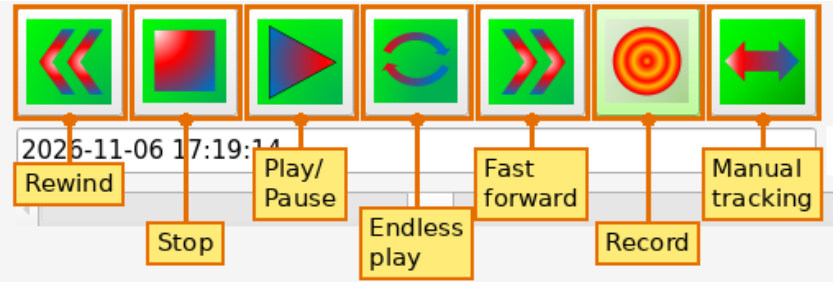


Fig. 1-8: Transport buttons of the player

The tracking button needs a little explanation: Sometimes it can happen that during playback the stations either sound very disturbed or even just like noise. This sometimes happens due to bit errors during data transmission. The spectra can also change considerably and frequency shifts can occur. Although this can often be 'repaired' by restarting the playback, it is easier to press the tracking button 1 - 3 times. The read counter then jumps by two bytes each time and the signal can be found again.

The small field 'BUF UFL' at the top right corner of the logo is normally invisible. It only appears (red) if a buffer underflow occurred during a recording, i.e. if data could not be transferred in time, and remains visible until the next recording is started.

Some devices need a while after Play before they deliver data. In this case a small window 'Starting SDR device' appears; it closes automatically as soon as playback runs, or when you press STOP. This

happens e.g. with an fl2k dongle that has only just been plugged in: it must first be released from the kernel mass storage driver, which may take more than 10 s. With the STEMLAB drivers the window appears while the SDR server on the STEMLAB is started and connected.

In the top right corner of the player there is the button 'CONFIGURATION ...'. It opens a separate window with the settings that are needed only rarely (Fig. *1-9*): the IP address of the STEMLAB, the startup tab, the recording path, the logfile and, under 'Display / test', the spectrum monitor and the test mode. The window looks the same in all skins, may stay open while the player is running and is hidden again with 'Close'. In skin 5 the spectrum monitor is located directly below the button. To change the IP address of the STEMLAB, click 'Set STEMLAB IP Address' in this window, edit the address and then save it.

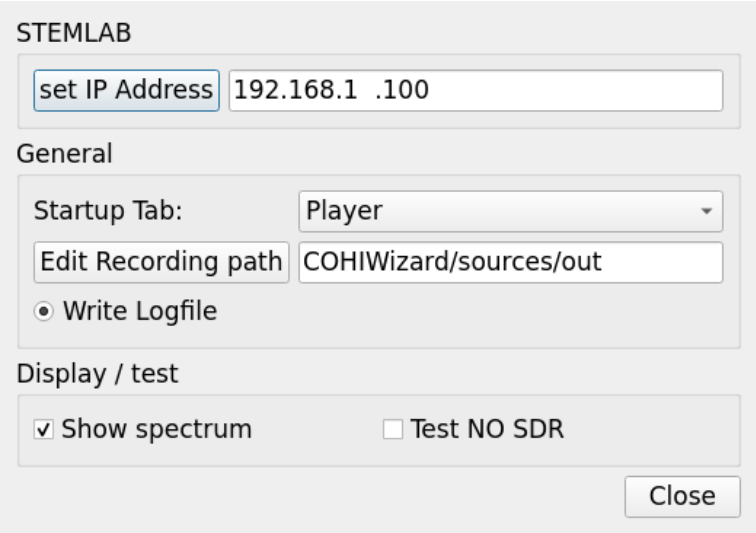


Fig. 1-9: The configuration dialog

The IP address is only needed for the STEMLAB drivers. With the USB devices (fl2k, ADALM2000) the field and the button are inactive.

When using Skin 2 or 3 the playerbutton group appears somewhat differently, see fig. *1-10* . All buttons are larger for more convenient use on a touchscreen and the tracking button here is located above the FF-button.



Fig. 1-10: Player button group under skin 2.

Since v2.2.0 the COHIWizard can automatically search for the address of a connected STEMLAB-device. When clicking 'Set STEMLAB address' the first time the operating system will ask you for administrator access, which you must grant. Then the COHIWizard will scan for IP addresses and report whether the search was successful. If so, the new address automatically appears in the address field and you just need to confirm it. Then press 'save IP address' to store it in the system configuration.

Should the identification procedure fail, please try once again. Especially at the first use this function may indeed fail. Should the problem persist, then you can still enter the address manually. **Caution:** If you only have to enter 2 instead of 3 digits in a field, you must explicitly delete the preconfigured 3rd digit; unfortunately, pressing 'Tab' will leave the old entries in place.

The configuration dialog also contains the drop-down menu ‘Startup Tab’. If you click on it, you get a list of the currently available tabs and can select the one that should be shown when the program starts. If you mostly use the player and therefore select ‘Player’, the program will always start in the player view. If you select ‘Resample’, for example, because you resample a lot, you can set this tab as the start view.

The button ‘Edit Recording path’ in the configuration dialog allows you to define the path for files to be recorded, if recording is supported by the chosen device. Recordings are then always saved there automatically. The same path is used as default output path of the resampler (see the respective section).

There is also a radio button ‘Write Logfile’ in the configuration dialog. If this is activated, a debug log file is written in the background, which is useful for analyzing program errors during further development of the software. This is usually unimportant for users of the software, but can be useful for communication between users and programmers during troubleshooting procedures. The setting is available in all skins.

The Radio button ‘Offset’ in the light-green field 'BAND PARAMETERS' allows you to set a frequency offset to the LO frequency (band center frequency) for playback. This allows you, for example, to play back an LW recording with a center frequency of 180 kHz in the medium wave range of a radio by activating an offset of e.g. 920 kHz. The new center frequency is then 1100 kHz. This can sometimes be useful if the band for a particular recording is not available on a particular receiver or is defective. An activation of the offset is indicated activating the radio button 'Offset (kHz)' (skins 2 and 3: 'Offset') being highlighted in yellow. Then enter the desired offset value (in kHz) in the edit field at the right. ATTENTION: This function does NOT affect the recorder (see next section), the LO offset is ignored during recordings.

Under skin 2 the LO offset edit field has been replaced by a spinbox, which allows to change values without keyboard/mouse by pressing the up/down arrows on a touchscreen (see fig. *1-11*). The values there can be changed between -10000 and + 10000 kHz.

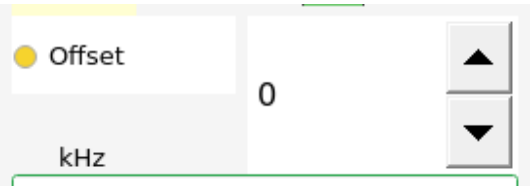


Fig. 1-11: LO offset spinbox under skin 2.

The playlist is located below the scroll bar: on the left the list of the files in the directory of the loaded file, on the right the playlist. It is inactive by default, but can be activated by pressing the 'Activate Playlist' button (under skins 2 and 3 the button then reads 'Select Files (drag)'). You can then drag and drop individual files from the left list into the 'Playlist' field. If you then press the play button, the playlist will be played. If you have also activated the 'Autorepeat' button, the playlist has priority over the repeat function, i.e. it is only repeated after the list has been played.

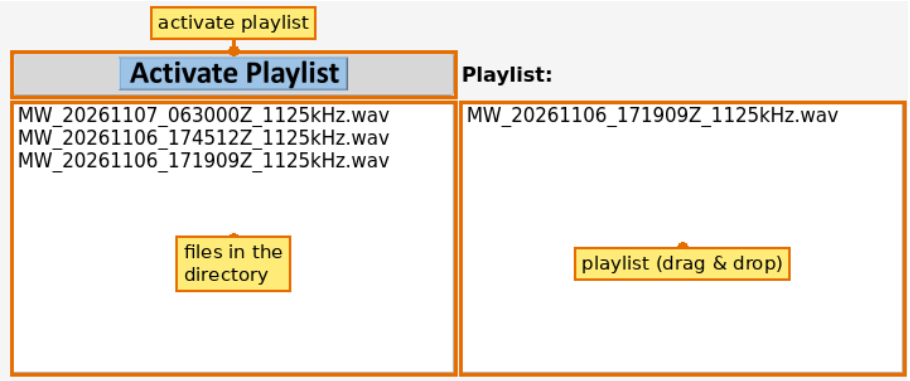


Fig. 1-12: Activated playlist: drag files from the left list into the playlist

However, if files are selected that have a successor file entered in the 'nextfile' header (typically for series of 2GB files from a longer recording), then this file series is played with priority before skipping to the next file in the playlist. An autorepeat is also subordinate to this automatic function. The order is always:

- Play all files of a series of related recordings

- Play the playlist

- Then return to the beginning with autorepeat

Automatic playback of a contiguous series also takes place if only one file in the series is selected and the playlist is not activated at all.

1.5.2 Modulator types and audio station lists

The universal drivers can generate AM carriers from audio sources themselves. What is sent is selected with the combo box 'MODE' (Fig. *1-13*):

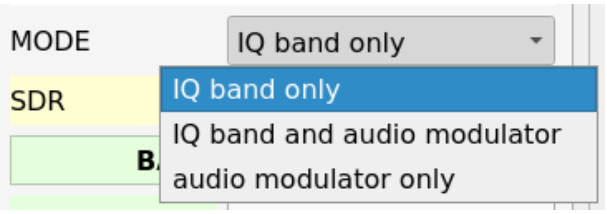


Fig. 1-13: Selection of the modulator type

- **IQ band only:** classic playback of an IQ file. All installed drivers are offered.
- **IQ band and audio modulator:** the IQ file is played back and the stations of an audio station list are added as additional AM carriers. Only the universal drivers are offered. The 'Mixer' slider is active in this mode.
- **audio modulator only:** no IQ file is needed, the band consists of the stations of the audio station list only. Only the universal drivers are offered.

In the two audio modes the fields 'BAND PARAMETERS' and 'REC timer' are inactive, as recording is not possible there. Switching back to 'IQ band only' unloads the station list. Under skins 2 and 3 the

entries are abbreviated: ‘IQB only’, ‘IQB & AUM’ and ‘AUM only’ (IQB = IQ band, AUM = audio modulator).

Audio station list: The stations are defined in a text file in CSV format, which is loaded via the menu ‘File → Playlist open’ (Alt+L), under skin 2 and 3 with the button ‘csv ...’ in the bottom line. Each line describes one carrier with four fields separated by semicolons: carrier frequency; bandwidth; programme name; source. Frequency and bandwidth can be given with the units Hz, kHz or MHz (a plain number means Hz); if the bandwidth is empty, 9 kHz is used. A header line and lines beginning with ‘#’ are ignored. Example:

```
Frequenz;Bandbreite;Programmname;URL
592 kHz;4.5 kHz;ORF Radio Wien;http://orf-live.ors-shoutcast.at/wie-q2a
610 kHz;4.5 kHz;UK 1940s Radio
Station;http://solid2.streamupsolutions.com:24929/stream
1008 kHz;4.5 kHz;My evergreens;music/evergreens.m3u
1179 kHz;4.5 kHz;Single title;music/title01.mp3
```

The source can be the URL of a web radio stream, a local audio file in any format readable by ffmpeg, or a playlist (m3u, m3u8, pls) with local files or URLs; the titles of a playlist are played one after the other. Relative paths in the CSV refer to the folder of the CSV file, relative entries in an m3u to the folder of the m3u file. ‘/’ and ‘\’ are both accepted. Hence a folder with the CSV file, the playlists and the music can be copied to any place or computer, even with another operating system, without editing. Web streams need an internet connection.

When you press Play, all stations that cannot be played (file or playlist not found, no playable entry, ffmpeg cannot be started) are listed in one info box; the remaining stations are played. Choose the carrier frequencies within the band of the device and, in the mixed mode, preferably in free channels of the recording.

Mixer: In the mode ‘IQ band and audio modulator’ the slider ‘Mixer’ sets the level of the audio carriers relative to the IQ band between −30 and +30 dB (Fig. *1-14*). In skin 5 it is the vertical slider next to the transport buttons, in skins 2 and 3 a horizontal slider below the SDR device selection. It acts immediately during playback, so that the stations from the list can be adjusted to the loudness of the recorded stations by ear. The position is stored and restored at the next start. ‘Gain’ and ‘AVC’ act on the sum of both signals. The base level of the audio carriers can be changed in config_wizard.yaml (section 3.3, Case 4).

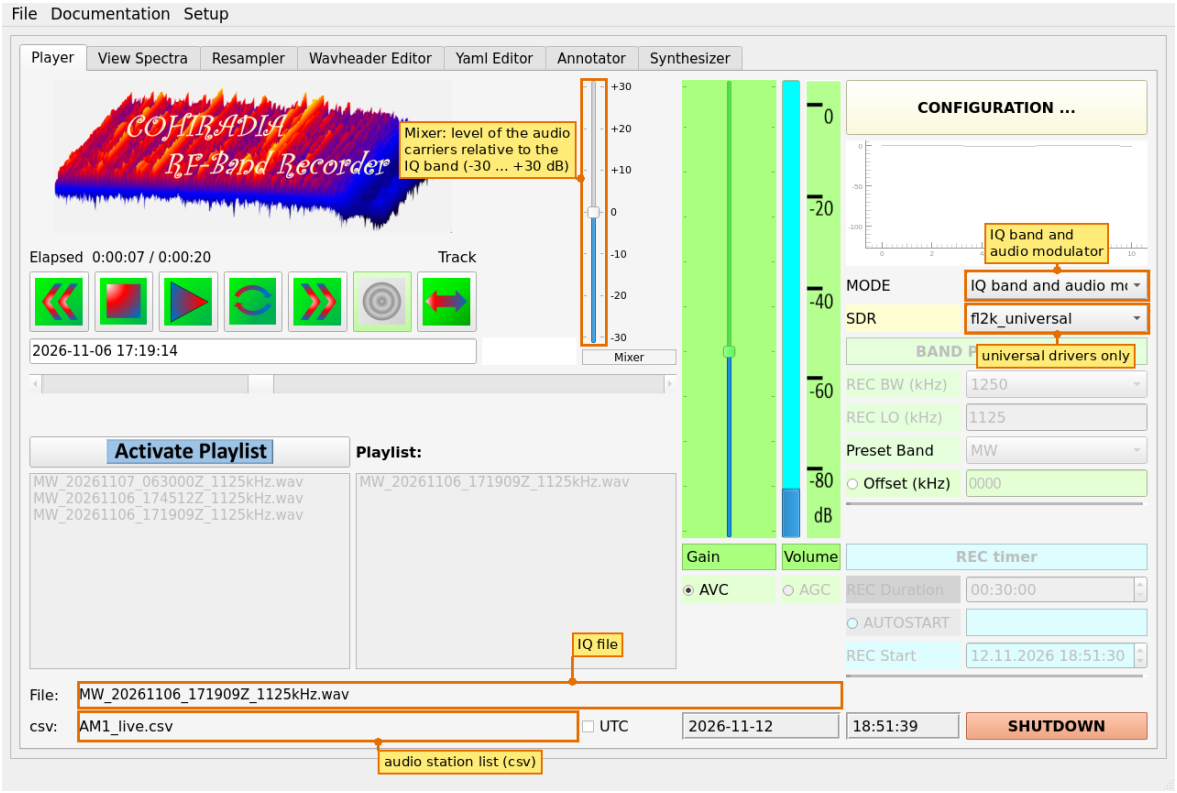


Fig. 1-14: Mode 'IQ band and audio modulator' with IQ file and audio station list loaded

Important remark: when using the IQ band and audio modulator there may occur strong interference tones around some carriers, frequently 1 kHz or multiples thereof. This may happen, if carrier frequencies of the csv-file do not align with the frequency raster used in the IQ-file (e.g. the European 9-kHz-raster). If you e.g. mix an artificial carrier of 764 kHz with an IQ-File, which contains a station at 765 kHz, then a typical 'beat'-tome at 1 kHz is audible, which may be very annoying. In this case change the carrier frequency in the csv-File to a free one, a multiple of 9 kHz away from 765.

1.5.3 Recorder (not available under skin 2 or 3 !)

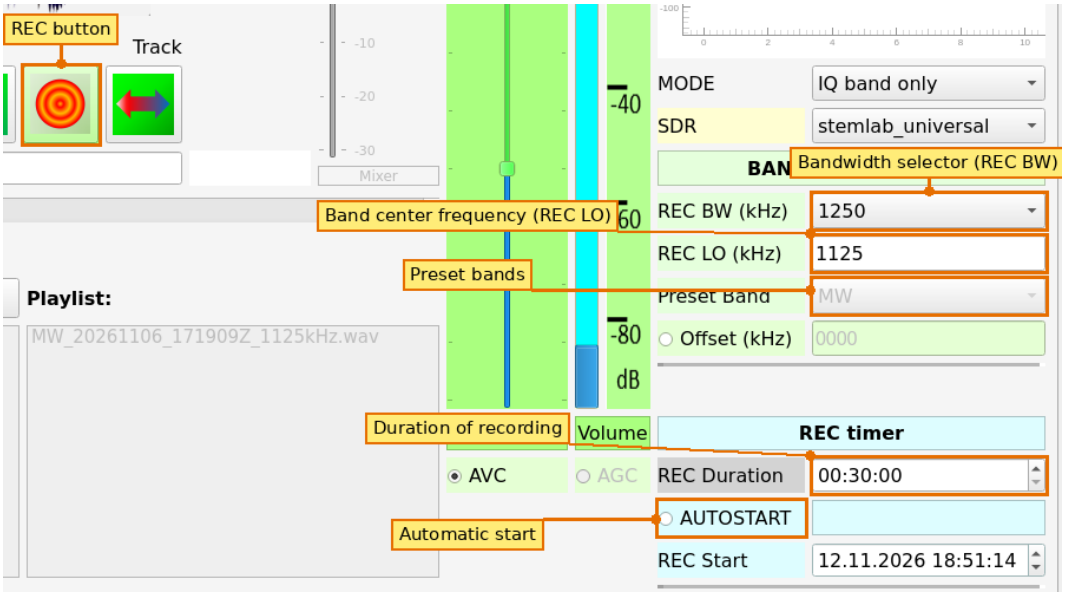


Fig. 1-15: Recorder and timer functions

Recording is only possible with the STEMLAB drivers (stemlab_universal, stemlab_125_14) and the modulator type ‘IQ band only’.

You can start your own recording by pressing the orange-red Rec button in the player button field. The recording only runs for a certain amount of time to prevent unintentional overflowing of the storage medium. This time is set in the light-blue 'TIMER' field at the bottom right (see 1-15). The default setting is 30 minutes, which must be adjusted for your own purposes. The desired recording duration must be entered in the format hh:mm:ss. Every recording, regardless of whether it is started manually or automatically by timer, ends after this time.

It should be noted here that recordings require a lot of memory, namely

Memory requirement (bytes) = 4 x bandwidth (Hz) x time (s).

For 30 minutes of MW recording you typically need 1250000 x 4 x 1800 = 9,000,000,000 bytes = 9 GBytes!

The other buttons in the 'REC timer' field allow you to set a time for automatic recording. If you activate the radio button 'AUTOSTART', the 'REC Start' field is highlighted in yellow and activated. A time is automatically entered there that is 15 minutes after the current system time. You now have 15 minutes to enter the desired recording time in the 'REC Start' field. However, the entire date must be set correctly in the form DD:MM:YYYY hh:mm:ss. Recording will then start automatically at the set time. Until then, a countdown will run in the time display field below the player button group (see Fig. 1-16).

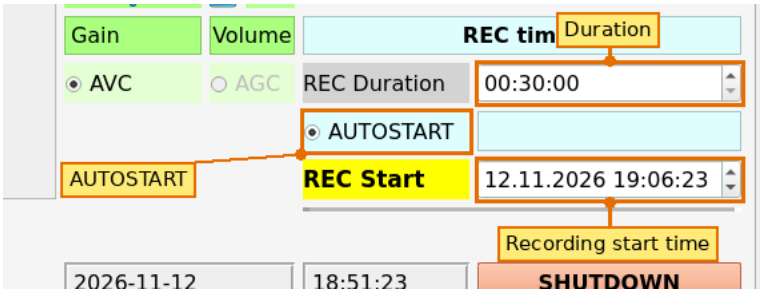


Fig. 1-16: Automatic start of a recording: REC AUTOSTART activated

The green 'Band Parameters' field allows you to set the recording parameters for recordings. These are the bandwidth (field ‘REC BW (kHz)’, 20 to 2500 kHz for STEMLAB) and the band center frequency (field ‘REC LO (kHz)’, LO or Local Oscillator) in kHz. The 'Preset Band' option allows you to conveniently select a number of preconfigured AM bands with their characteristic settings for bandwidth and center frequency.

During a recording, the logarithmic 'Volume' display serves as an indicator for the level of the STEMLAB. You should select the input signal level so that you do not overdrive under any circumstances, but also make good use of the SDR's dynamic range. A level of up to max. -15dB (volts), preferably -20dB, but preferably not below -40dB, has proven to provide good intermodulation resistance. The 'Gain' control is not intended for recordings, as the STEMLAB has no adjustable preamplifier. Suitable preamplifiers are therefore required for recordings from an antenna. True AGC is not possible and AVC as well as the Gain control slider are inactivated by default.

However, since v2.0 there is an (experimental) possibility for virtual manual and automatic gain control. As this is not an feature in v2.0.0, the respective function needs to be activated manually in the file COHI_root/sources/config_wizard.yaml. Open this file with a Text-Editor. There is a line

REC_AGC: false. Change this line to REC_AGC: true. When re-starting the COHIWizard, then the AVC button will also be available in the recording mode.

In this mode the numerical data values in the IQ-files will be normalized so that they fill the dynamic range up to a certain threshold value (ideally 80% of FSR). However, this does not prevent clipping in case the input signal exceeds the +/- 1V limit of the STEMLAB, nor does it increase the resolution. Hence this function only serves for maintaining a fairly constant signal in the IQ-file and is thus of very limited usefulness.

In that respect the AGC is similar to digital zoom in a camera: it just resizes the image without changing the resolution.

1.5.4 Special functions

In the configuration dialog ('Display / test') there is the special function 'Test NO SDR', which you may tick for test purposes if you want to test the player without having an SDR connected. This function is normally only required when developing programs and is therefore of no interest to most users.

The checkbox 'Show spectrum' in the configuration dialog switches the spectrum monitor on and off (skin 5: below the button 'CONFIGURATION ...', skins 2 and 3: bottom right). Switching it off reduces the computing load on slow computers.

In the bottom line of the GUI there are the time display elements on the right with the option to switch the displayed time to UTC and the SHUTDOWN-button. It is recommended to shut down STEMLAB's LINUX at the end of a session by pressing that button before turning off the power; you will receive a pop-up message as soon as the system is ready to be switched off.

Under skin 2 and 3, only the SHUTDOWN-button is available: under skin 2 below the volume control group, under skin 3 in the right column below the LO offset.

1.5.5 Device drivers

Since v2.3 every device is operated by a driver in the folder sources/dev_drivers. Two families are available; which of them are installed is chosen in the setup tool (section 1.2):

- **Universal drivers (recommended):** fl2k_universal, stemlab_universal and ADALM2000_universal. They support all three modulator types (section 1.5.2): IQ band, IQ band plus audio carriers and audio carriers only. The computationally expensive parts (upsampling, synthesis of the audio carriers, mixing) run in C++ libraries (dev_drivers/libs, ADALM2000_universal) instead of an ffmpeg pipeline.
- **Reference drivers:** fl2k_stream, stemlab_125_14 and ADALM2000. These are the drivers of v2.2. They only play IQ files ('IQ band only') and are kept for comparison and as a fallback.

Recording is possible with the STEMLAB drivers only (stemlab_universal and stemlab_125_14) in the mode 'IQ band only'.

1.5.5.1 STEMLAB 125-14

This is the originally supported device driver which has been tested excessively in the past. It can be used for both the STEMLAB 125-14 and the STEMLAB 125-10. The latter is the 10-bit version which is significantly cheaper and may be sufficient for playback of most AM spectra. It has, however, not been

tested very much and may provide slightly worse audio quality in case of recordings with large dynamic range and low background noise, like those produced by the synthesizer tool.

Selecting this device will send the signal to a STEMLAB 125 if connected via LAN or WLAN.

With stemlab_universal the audio carriers are synthesized on the PC and added to the IQ stream before it is sent to the STEMLAB, so that the STEMLAB itself needs no additional software.

1.5.5.2 Fl2k USB-VGA Dongle

This driver enables playback via low-cost USB-VGA converters based on the Fresco Logic FL2000 chip, as described on the osmo-fl2k webpage. This device has three 8-bit DACs with sampling rates between 10 and 100 MS/s. There are several projects in the Web which are dedicated to the generation of RF-signals, among those also wideband transmission of AM spectra (see e.g. the one on radiobastler.de).

With 8 bits the playback quality is inferior to that of a STEMLAB, especially with signals with large dynamic range / good SNR. Nevertheless this device may be sufficient for many standard applications and extremely cheap.

For detailed descriptions of the hardware and the pinning of the output ports see the osmo-fl2k webpage. Under Windows the dongle needs the WinUSB driver, which is installed with the tool Zadig. The setup tool offers to download and start Zadig when an fl2k driver is installed for the first time, and the player offers it again if the dongle cannot be opened. In Zadig select 'Options → List All Devices', choose the device with the USB ID 1D5C 2000, select 'WinUSB' and press 'Install Driver' (or 'Replace Driver'). Under LINUX the setup tool installs the osmo-fl2k packages and a udev rule, so that the dongle can be used without root rights. If the dongle is plugged in only while COHIWizard is running, the first start of playback can take more than 10 s; the window 'Starting SDR device' is shown meanwhile (section 1.5.1).

As the dongle does not upsample from the baseband to the target RF band itself, this is done on the PC. The universal driver fl2k_universal uses an efficient polyphase filter in a C++ library and plays MW, LW and SW recordings reliably on current PCs. The following limitations were found with the reference driver fl2k_stream, which uses an ffmpeg pipeline; they are kept here for users of that driver:

Under Windows the optimum situation is if the ratio between the sampling rate (SR) of the dongle (an integer multiple of 10MS/s) and the SR of the IQ-file is an integer power of 2, i.e. 2^N. Under this condition signals up to 5MHz can usually be played back without problems. Therefore e.g. MW-Files with 1250 kS/s and 16BpS typically are well suited.

With sampling rates which do not obey the above rule (e.g. 500kS/s), *ffmpeg* may slow down and the pipeline may block/crash. Higher resolutions than 16 bit (24 and 32 fps) have also been found to be problematic.

However, if the SR is low enough (e.g. 250 kS/s for LW), problems practically never occur, even if it does not obey the power-of-2-rule.

The problems can be reduced by switching off the spectral monitoring, but then the volume display no longer works, so you can only adjust the gain when 'flying blind'.

SW recordings with LO around 6kHz (i.e. target sampling rate 20MS/s) often come to a standstill, especially when reading from an external hard disc via a USB hub to which the fl2k dongle is also connected. When reading from the internal SSD, you can usually also play back SW recordings with LO around 6kHz (i.e. target sampling rate 20MS/s), provided the original SR is 250 or 500 kS/s. The SR of

333.3kS/s used by the RSP1a/SDRUno on the 49m band is very unfavourable and regularly leads to problems. These recordings should therefore always be resampled to e.g. 500kS/s or 250kS/s (by truncating the band).

Under LINUX, the most favourable mode (SR = 500 kS/s, LO = 6050 kHz, internal SSD, spectrum monitor switched off) has not yet led to any crashes; under Windows, the system usually gets stuck after some time of playback and has to be restarted.

Conclusion: MW and LW can usually be played back without any problems, but there are considerable restrictions when trying to use sampling rates which are not integer fractions or multiples of 10 MS/s.

Tests have been carried on the following PC:

LENOVO, model 20N4002UGE
CPU: Intel64 Family 6 Model 142 Stepping 11 GenuineIntel ~1792 MHz
Physical memory: 40 708 MB
Available memory: 25 540 MB
Virtual memory: max 46 596 MB
OS: Microsoft Windows 10 Pro Education N10.0.19045
OS: LINUX Debian13

1.5.5.3 ADALM2000

General:

This driver allows the playback of IQ files with upper cut-off frequencies up to 2.5 MHz on the ADALM2000. This USB-powered module is considerably cheaper than a STEMLAB125-14 and comes with 2 ADC and 2 DAC channels (12-bit resolution each), and a GPIO port. It can be operated as an oscilloscope, function generator or logic analyser and, together with the new COHIWizard, allows the playback of COHIRADIA IQ files. However, there are limitations: As with the fl2k, the upconversion to the target band is not done on board but on the PC. This requires considerable computation power. Additionally, due to the use of USB2.0 and the used data format, the transfer rate is limited. In order to prevent packet delays and periodic interruptions of the playback stream, the allowable signal frequency in the RF-bands was limited to 2500 kHz. In any case the computation power of the PC has to be high enough, otherwise also interruptions can happen. The band limit is o.k. for LW and MW while SW recordings can only be played back with a trick: Setting the option 'LO offset' to a suitable value (say -4000 kHz for the 49m band) the signals can be down-converted and played back in the medium wave band of the radio. Initial tests showed usable playback for MW and LW with better dynamic range than the fl2k.

Important note: Unfortunately the ADALM2000 does not possess good antialiasing filters at its DAC-output. Therefore aliases of the useful band appear in the specrum. Though they appear above the nominal end of the band played back, they may cause spurious noise in the radio receivers, yielding an apparently much worse SNR that one would expect from 12 bits. Thus it is highly recommended to place an external lowpass between the Output of the ADALM2000 and the receivers which should be fed by the signal. The cutoff frequency should be around 1.7 – 1.8 MHz (end of the medium wave band). Good experience exists with a passive Butterworth filter of 4th order.

If there are still perturbations like strong noise when playing back certain recordings, sometimes a manual adjustment of some parameters may be necessary in the config_wizard.yaml File. See also the section on manually adjusting these parameters.

Important note: The drivers need the libm2k library. The setup tool installs it automatically when an ADALM2000 driver is selected (LINUX x86_64: Python wheel with all native libraries; Windows: libm2k DLLs from the official release and optionally the ADI USB drivers). Only if this fails, install libm2k manually from https://github.com/analogdevicesinc/libm2k/releases (under Windows the full installer libm2k-*.exe, not only the Python bindings).

1.6 TAB ‚VIEW SPECTRA': VISUAL SIGNAL CHECK OF RECORDINGS

In the 'View Spectra' tab, you can scan through the recording with a scroll bar and display the spectrum at different points in time. To do this, a wav file with the SDRUno format (SDR standard IQ data file) or a raw STEMLAB 'dat' file (headerless raw IQ file) must have been opened. The file name is displayed at the bottom of the window. The spectrum is updated at the corresponding time point when the scroll bar button is released. As shown in *Fig. 1-17*, you can zoom in via the tool bar above the graphic (magnifying glass), zoom/pan with the mouse buttons (cross symbol for pan/zoom) and change some graphic properties. It is also possible to save the graphic in a separate file.

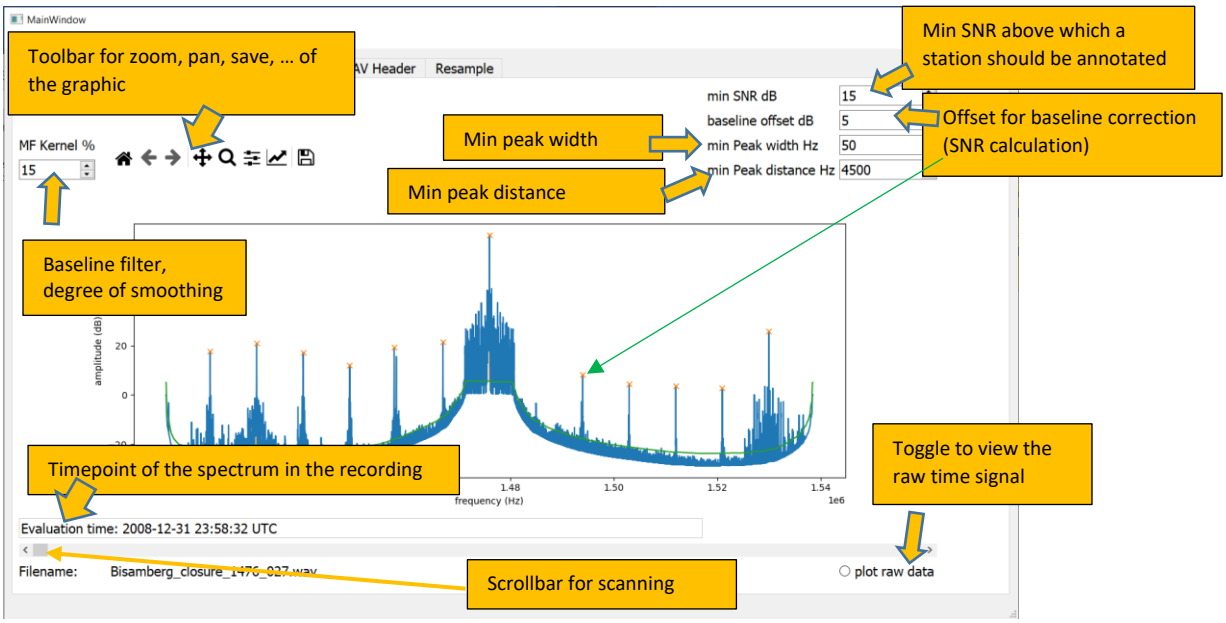


Fig. 1-17: Viewing the spectrum of a recording.

The time point of the spectrum is displayed in the gray field below the graphic in UTC, but is only correct in absolute terms if it stems from a wav file. For dat files, only the time difference relative to the start is displayed.

If you move the mouse pointer over individual peaks of the spectrum, the coordinates of the mouse pointer are displayed so that you can roughly 'measure' the spectrum. With the scroll bar above the 'Filename' display field you can scan through the recording, but you have to use the button of the scroll bar, clicking on > or < unfortunately does not work (at present).

The two spin boxes 'min SNR' and 'baseline offset' are not yet of central importance in this window, but are important later if you wish to annotate the file. The 'SNR' value is calculated automatically as the difference between a carrier peak and the green baseline drawn in the spectrum, as shown in Fig.

1-19. The values should therefore already be set here: 'baseline offset' can be used to shift the green baseline up/down if it does not fit well (see Fig. 1-18). You can use the arrow keys on the spin boxes to jump in 5 dB steps; for intermediate values, you must enter the corresponding numbers in the number field.

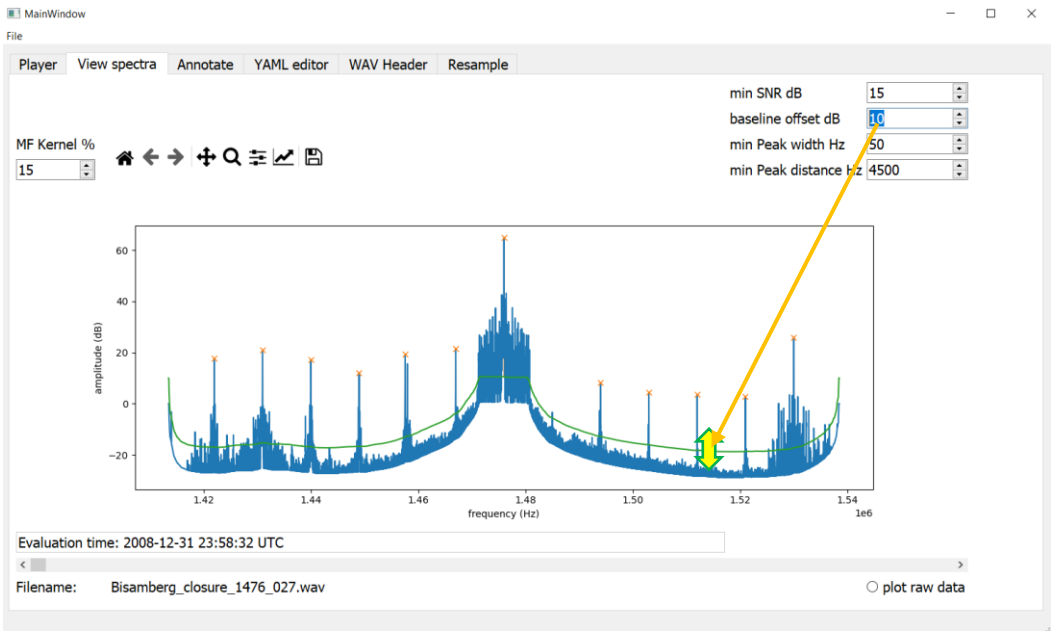


Fig. 1-18: Shifting the baseline for the SNR calculation.

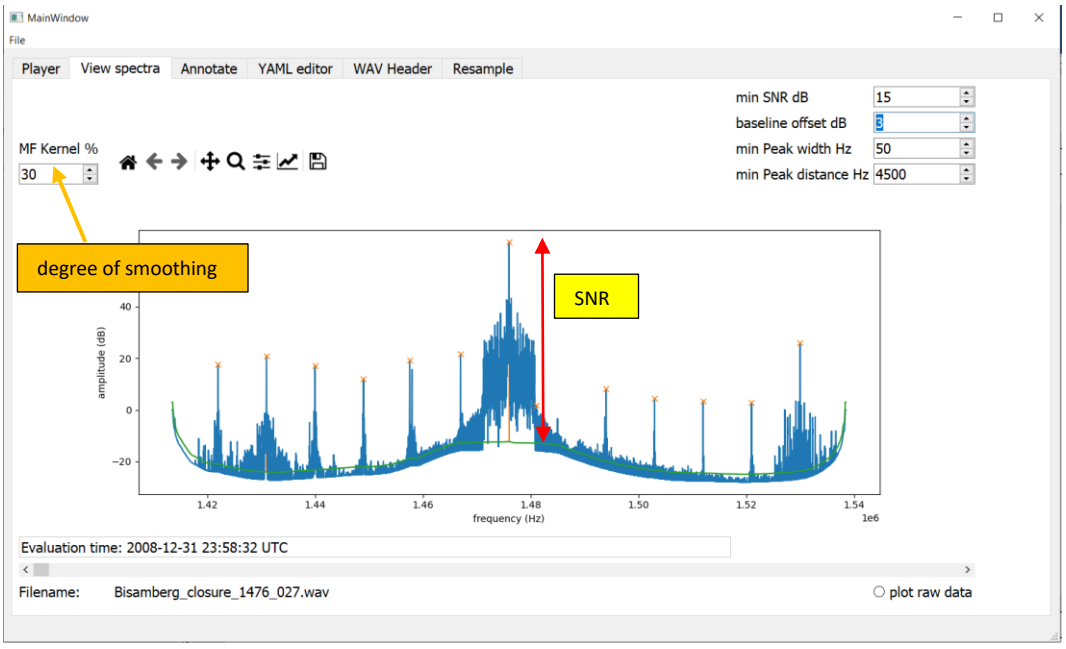


Fig. 1-19 Changed filter kernel (baseline smoothing), changed baseline, Definition of 'SNR'

The first suggestion for the baseline is determined from the spectrum using a smoothing filter. Ideally, the line should represent the noise level, but depending on the data set loaded, it may be slightly too high or too low and/or follow the peaks too closely, as in Fig. 1-18. In this case, it can be shifted up and down with 'baseline offset' and the degree of smoothing can be adjusted with 'MF-kernel'. The comparison is shown in Fig. 1-18 and Fig. 1-19.

In the 'min SNR' spin box, set the threshold value above which a peak is to be identified as a transmitter during annotation. IMPORTANT: Peaks below this value are ignored by the annotation tool.

The two spin boxes 'min Peak width' and 'min Peak distance' define the minimum width and the minimum distance to the neighbors a peak should have. However, values below 1 Hz and 100 Hz, respectively, cannot be set. The default settings are 50 Hz and 4500. This means that long-wave, medium-wave and short-wave radio patterns can normally be covered without any problems. Each time one of these two values is changed, the graphic is updated so that you can see what effect the setting has on peak detection: Peaks recognized as valid are marked with an orange cross.

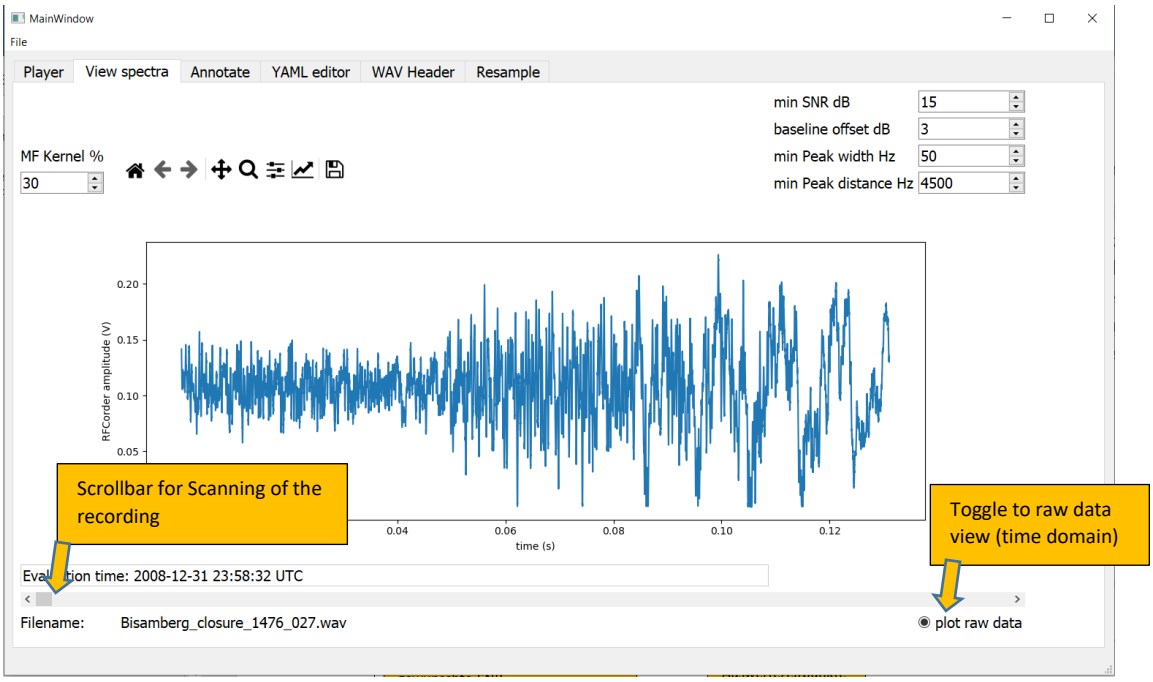


Fig. 1-20: Display of raw data.

If the radio button 'plot raw data' is activated, a section of the time signal is displayed, as shown in Fig. 1-20. This view is used to check for any clipping effects (if the signal exceeds the value 1) or, conversely, whether the signal is very weak and should possibly be amplified. This post-amplification can be of interest in the case of resampling, for example, and can be checked here.

The scroll bar below the window 'evaluation time' is also used in this mode to 'jump' through the recording, allowing you to scan through the file over time.

Please note: If the 'Manual Gain (dB)' field in the 'Resample' tab (see section 1.8) is set to a value other than zero, the spectrum shifts up or down on the ordinate axis. If the raw data is displayed, its amplitude increases/decreases according to the set gain. This function is used for level control when resampling.

1.7 TAB ‚ANNOTATE': IDENTIFICATION OF STATIONS AND ANNOTATION

After loading a file, you can automatically search for AM-carriers here and assign information on the corresponding stations to the peaks. The control elements of this tab are only enabled if a valid wav file has been loaded and the annotation for this file has not already been completed at an earlier time. Fig. 1-21 shows the operating window with some important setting elements.

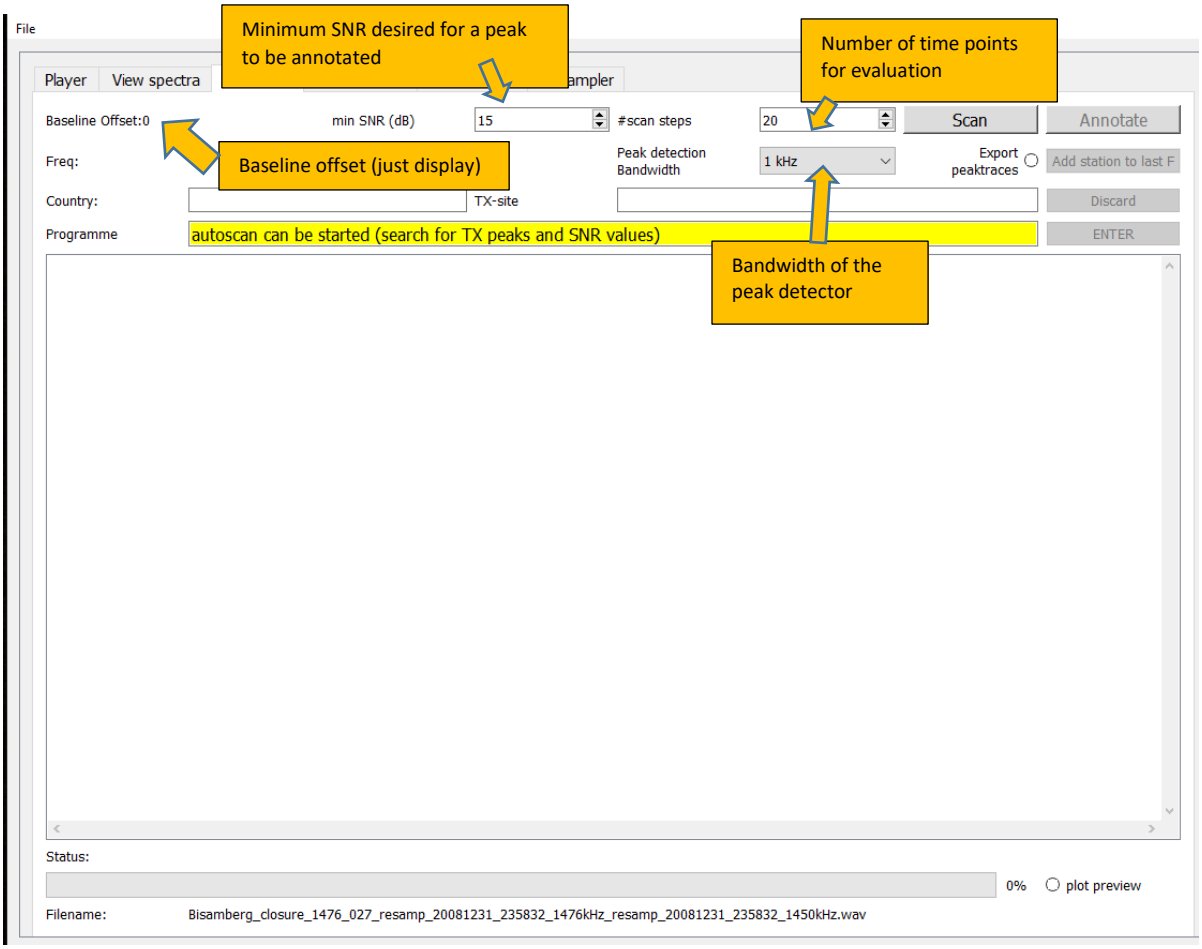


Fig. 1-21: Tab ‚Annotation' with the elements for setting relevant parameters.

Before annotating, you should check the values of the two spin boxes 'min SNR' and '#scan steps' and adjust them if necessary. 'min SNR' has the same function as in the 'Scanner'-tab, so can be set here as there. '#scan steps' determines the number of (equally spaced) points in time at which the recording is to be evaluated with regard to carrier peaks; the default setting is 20 points. The baseline offset is only displayed here. If you have not already done so, you should check in the 'Scanner'-tab whether the baseline is set satisfactorily (see notes in section 1.6). The evaluation is carried out in such a way that peaks that reach the minimum SNR (here 15dB) at least once at the selected times are considered valid. **IMPORTANT:** The SNR for the entry in the annotation yaml file is averaged over all evaluation time points in the current version.

Programmer's note: If this is not considered adequate, it may be changed in future versions, a selection option is also conceivable in principle, e.g. 'max SNR', 'min SNR' etc.). For helpful hints please contact the author.

The spin box 'Peak Detection Bandwidth' defines the resolution with which separate peaks are detected. The default setting here is 1 kHz, which is practically always suitable for the annotation of radio stations on MW and LW, as the frequencies are integer kHz values. However, if you want to annotate e.g. time signal stations or even peaks in the VLF band, the transmission frequencies are often not integer in kHz, but e.g. 129.1 or similar. The value would then be rounded to the nearest whole kHz value, which leads to incorrect entries (you would then have to edit the yaml files manually). If you want to prevent this, you can also set 100Hz or even 10 Hz, which leads to a correct display up to the first or second decimal place of the kHz specification.

WARNING: The options for 10 and 100 Hz have not yet been tested excessively, so they may still lead to unexpected SNR values. These settings should be regarded as experimental and hence be used with care.

Now you should also specify whether you would like to see brief plots of the spectra at the scan times during the scan. To do this, you can activate the radio button 'plot preview' at the bottom right of the window. This function only serves for diagnostic purposes and is hence switched off by default to enable faster scans.

For advanced users, the 'Export peaktraces' function is also available. This allows the amplitudes of the carriers found at the set times to be output to an Excel '.xlsx' file. This is an experimental and not extensively tested additional function that is not relevant for most users and is therefore not documented in detail.

The annotation itself takes place in 2 major steps:

1.7.1 Step 1: ‚Scan': Automatic identification of carrier peaks

As soon as you press the 'Scan' button, the program runs through the recording with the number of set steps and identifies the peaks. The progress is indicated by the status bar.

A file dialog window then opens and asks which reference station list you would like to use. If you have already annotated once, the last list used is automatically pre-selected by default. You can now select from a list of existing *.xmls files as shown in Fig. 1-22. Currently, only a European/African list is fully available, experimentally also one for New Zealand and surroundings and one for USA. These lists are supplied together with the installation files.

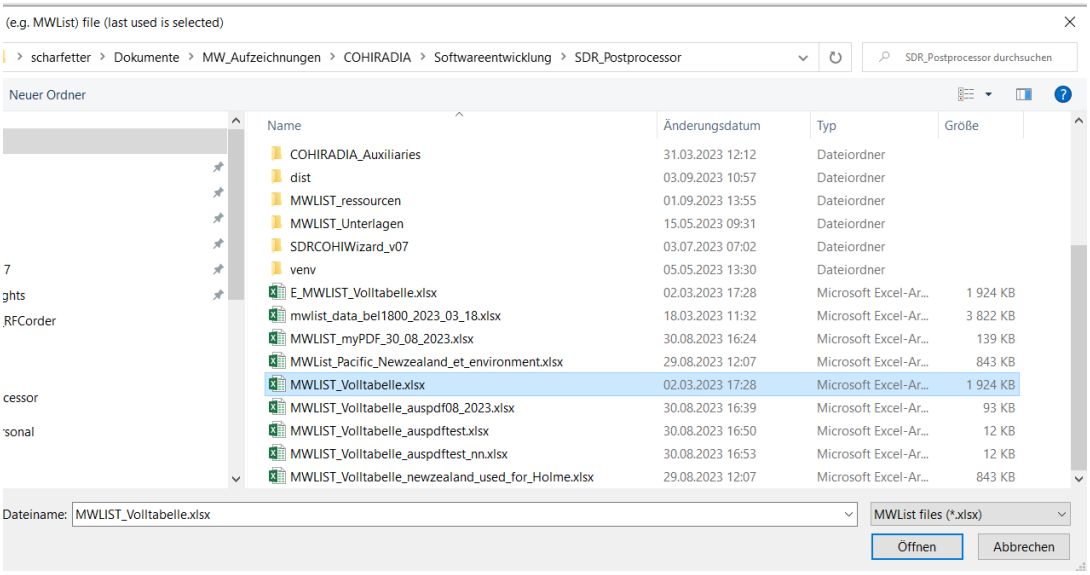


Fig. 1-22: Quest for choosing a stations list

After selection and confirmation, the reference station list is read and then the candidate stations for each carrier peak are saved in an auxiliary file (this file can be found in the COHI_root/ANN_[filename]/stations_list.yaml folder, if you are interested). As soon as this process is accomplished, the line ‘Programme’ turns yellow and displays the information 'autoscan has been completed, peaks and SNRs identified' (see Fig. 1-23). Furthermore, the 'Scan' button turns green and the 'Annotate' button is enabled.

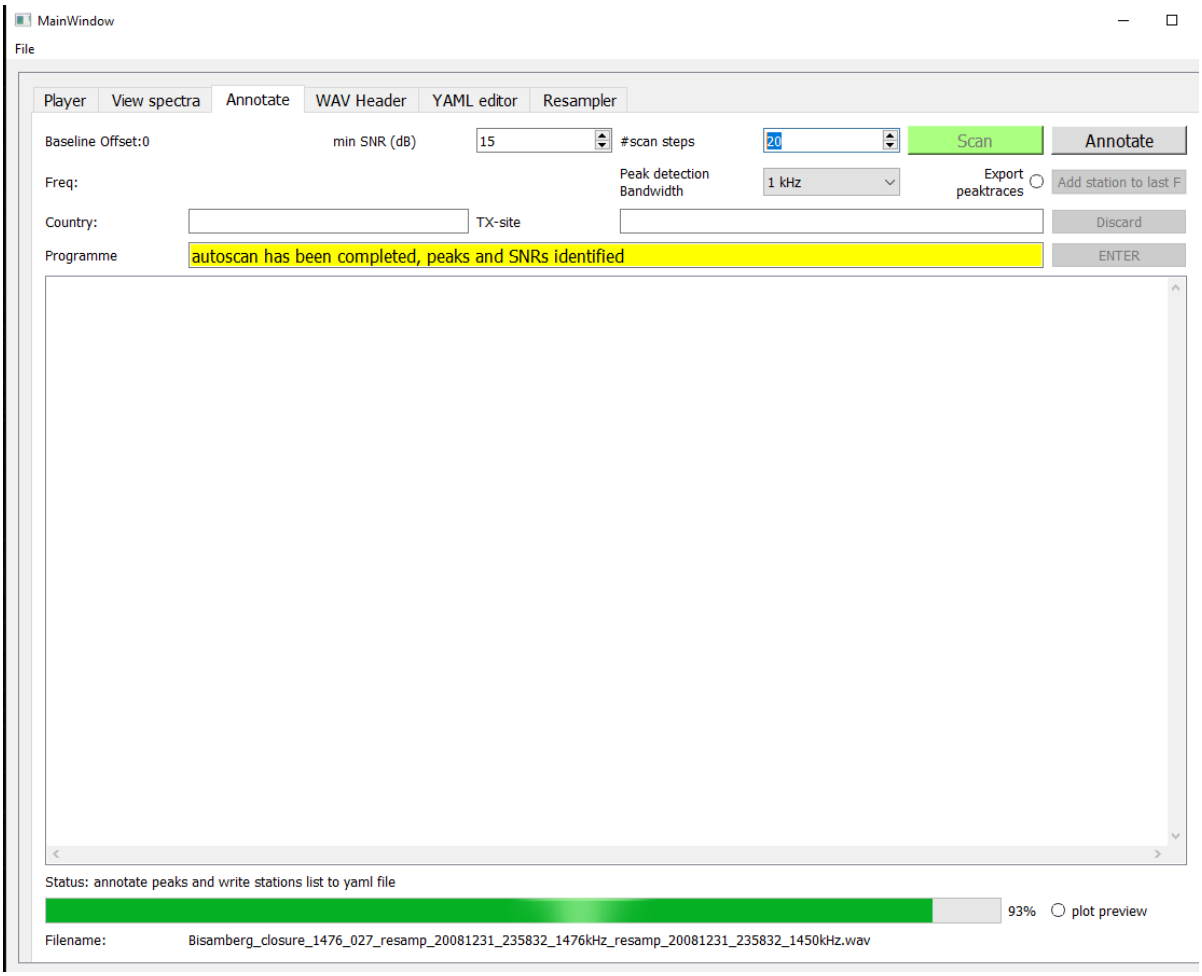


Fig. 1-23: Status after generation of the candidate station's list.

Please note: If you exit the selection of the reference station list file by clicking 'Cancel', you cannot continue with the annotation. You must reload the recording file (File open) and restart the annotation.

1.7.2 Step 2: ‚Annotate': Assignment of the correct stations

If you click on 'Annotate', the list of candidate stations for the next peak that has not yet been annotated is displayed. Ideally, it should look like Fig. 1-24.

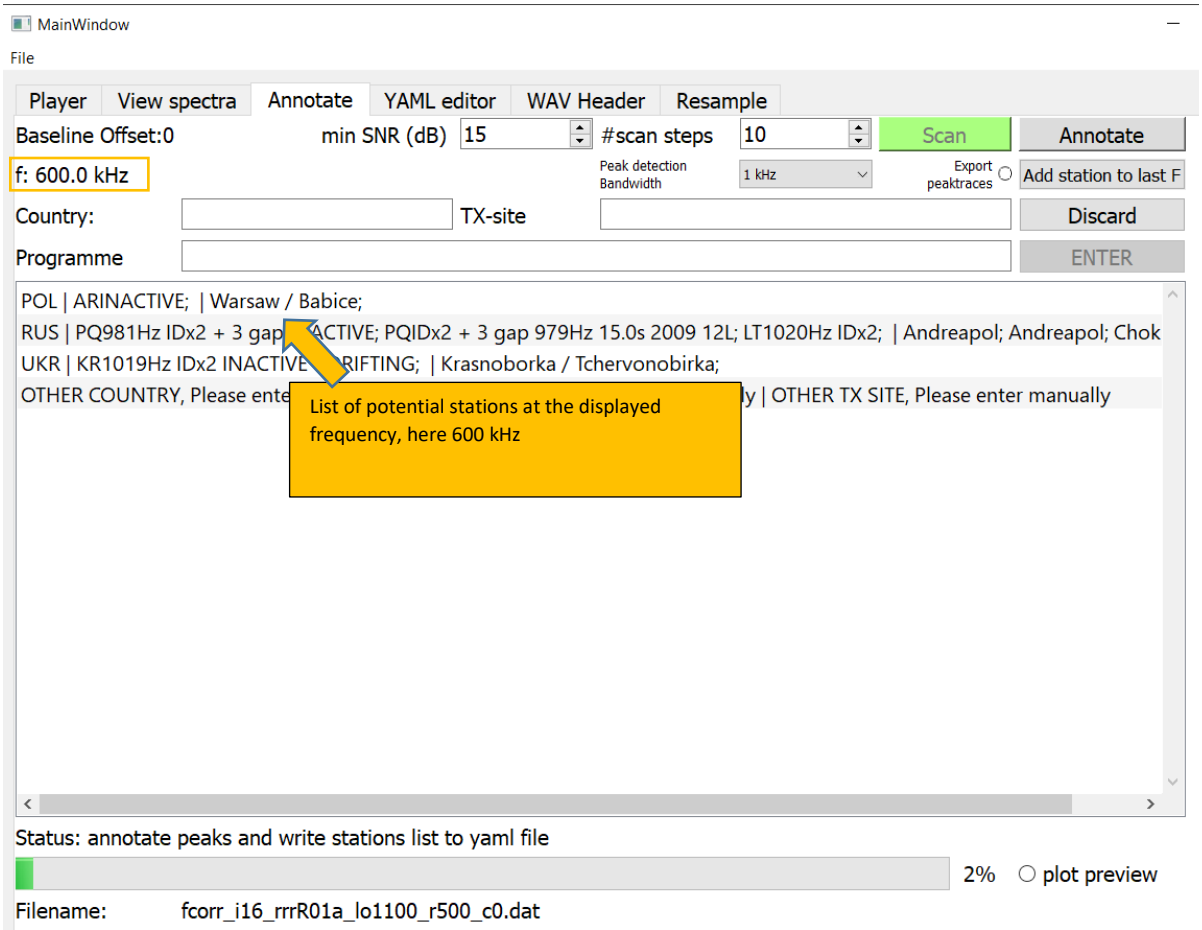


Fig. 1-24: List of the recommended stations at the displayed frequency

However, if no entry was found in the list of stations for the displayed frequency, only a line with 'not identified | not identified | not identified' appears, see Fig. 1-25. An annotation must be entered manually in the yaml file if it is a known signal.

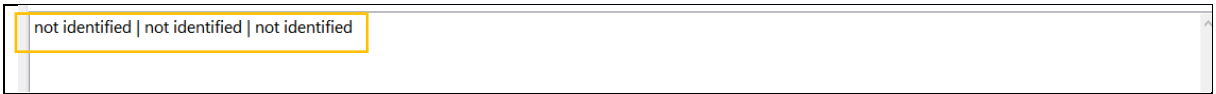


Fig. 1-25: Display of a unidentified carrier

If one of the displayed stations is correct, you can click on it to select it. The name of the selected station, the country and the transmitter site are now shown in the display fields within the orange marked area in Fig. 1-26, next to the frequency. The texts in these display fields can be edited (often the suggestions are still in a very rough form, possibly even incorrect), see Fig. 1-27. If you cannot find the actual station in the list, you can select the last option offered, 'OTHER COUNTRY', and then edit the entries manually with the correct details.

If you have clicked on the wrong station by mistake, you can reselect it at any time. If you do not want to accept a frequency at all (e. g. because you know that there is no carrier), you can click on the 'Discard' button without selecting one of the stations offered. Otherwise, confirm the entry with 'Enter' and it will be transferred to the yaml file. This can be done either by clicking on the button or by pressing the 'Enter' key on the PC. The program then jumps to the next peak. Alternatively, 'Discard' can also be triggered by the shortcut 'Alt →' (Alt - arrow key on the right).

Another feature is the 'Add station to last F' button. As it can happen that two stations transmitting on the same frequency appear at the same time (because both can be received at the recording site), i.e. they overlap, you can create a second entry on the same frequency. To do this, after entering the first station and then pressing 'Enter', press the 'Add station to last F' button. The annotator then jumps back to the frequency just used and allows a further entry. This can be repeated as often as required. The button can also be triggered with the shortcut 'Alt ←' ('Alt - left arrow key')

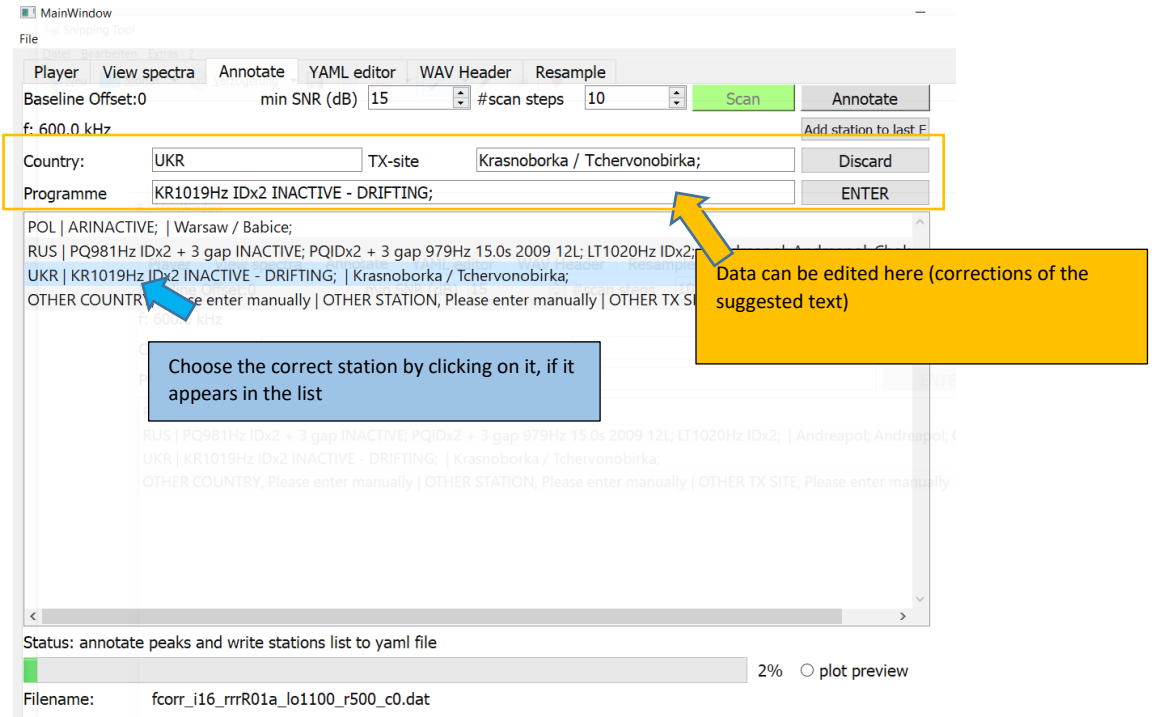


Fig. 1-26: Selection of a station name from the offered list of candidates.

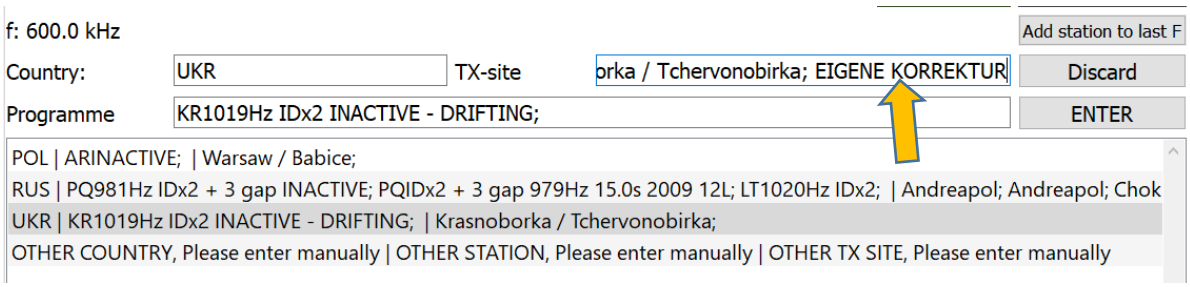


Fig. 1-27: Annotation after editing. In this example the text ‚EIGENE KORREKTUR' has been added.

Important remark: *The automatically suggested stations may well contain unwanted text or even be incorrect. You should therefore always check the information by listening to the recorded stations. The information comes from tables that were exported manually from 'mwlist.org' in 2022 and may no longer be up to date. In principle, each of the lists can also be corrected by the user if necessary by editing the Excel file.*

When you have finished annotating the entire frequency list (status is displayed on the status bar), the following is highlighted in yellow in the line next to the frequency: 'Record has already been annotated. For re-annotation delete annotation folder', see Fig. 1-28. The 'Scan', 'Annotate', 'Enter' and 'Discard' buttons are deactivated. 'Scan' and 'Annotate' now both appear green.

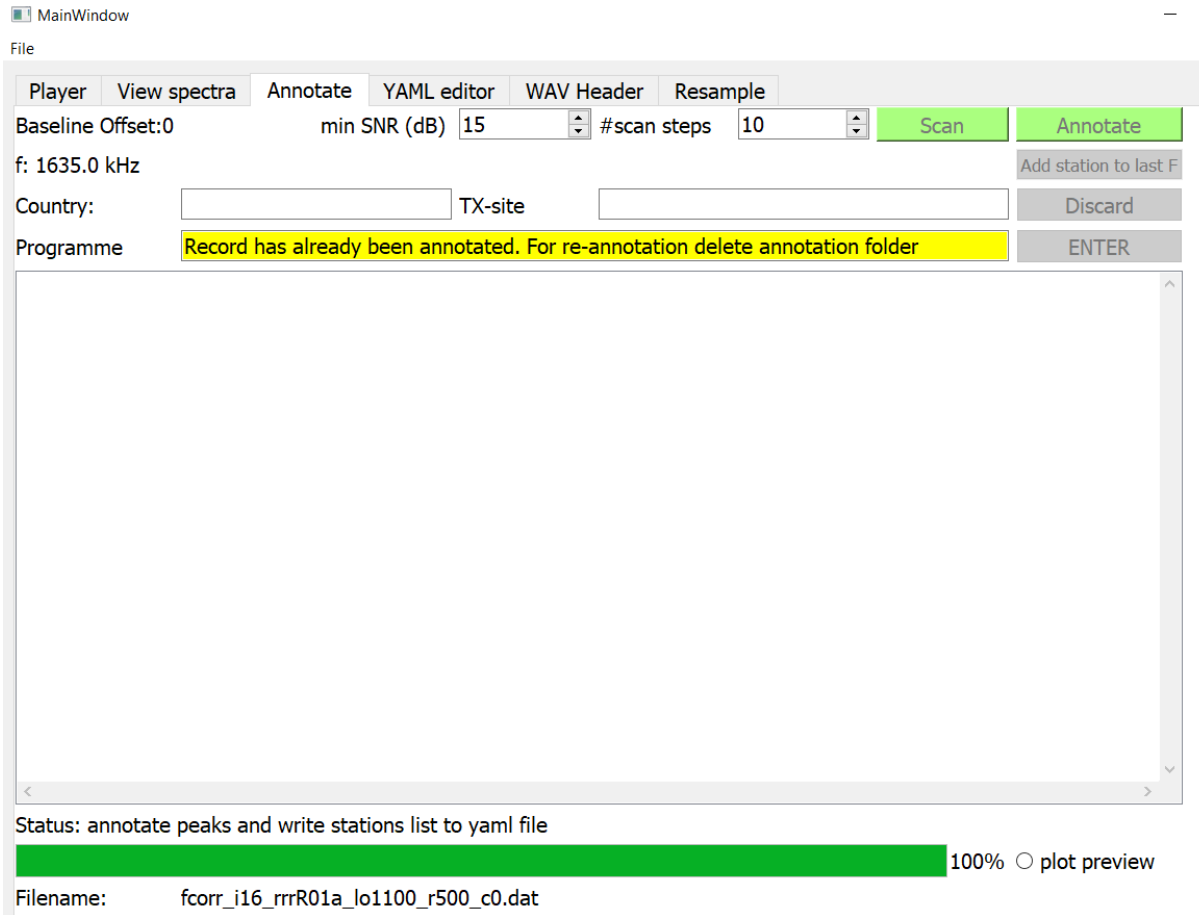


Fig. 1-28: Annotator view after accomplished annotation.

If you are not finished with the annotation in one session, you can close the program at any time and continue at another time. The program remembers the point at which you interrupted and continues from there after restarting.

Hint for experienced users: Unfortunately, there is currently no built-in function that allows you to jump backwards in the list that has already been created, except for a single frequency for double annotation. If you have skipped a station by mistake, you have to enter it manually into the yaml file later. Alternatively you can exit the program, open the file COHI_root/ANN_[filename]/status.yaml with a text editor, enter a correspondingly lower value (minimum 0) in the field 'freqindex' and restart the program. This option should be used with caution, as incorrect double entries can occur, which must be corrected manually.

1.8 TAB ‚WAV HEADER‘:

This tab is used to view and edit the wav header of a recording, if necessary. It can also be used to insert a wav header into a dat file which you have generated with the RFCorder, so as to create a valid wav file.

3 tables display the information in the header after loading a wav file and allow it to be edited if necessary and written back into the file. Since programs such as SDRUno or SDR# read this information, you must be very careful when making changes and know what you are doing. Otherwise the playback of the file may no longer work. For this reason, you must explicitly enable editing of the table with the 'EDIT' radio button.

If you then want to overwrite the existing wav header with the new data, press the button 'Overwrite Header' next to the 'EDIT' radio button (it is only active in EDIT mode), see Fig. 1-29. 'Overwrite header' can also be triggered by the shortcut ‘Alt-H’.

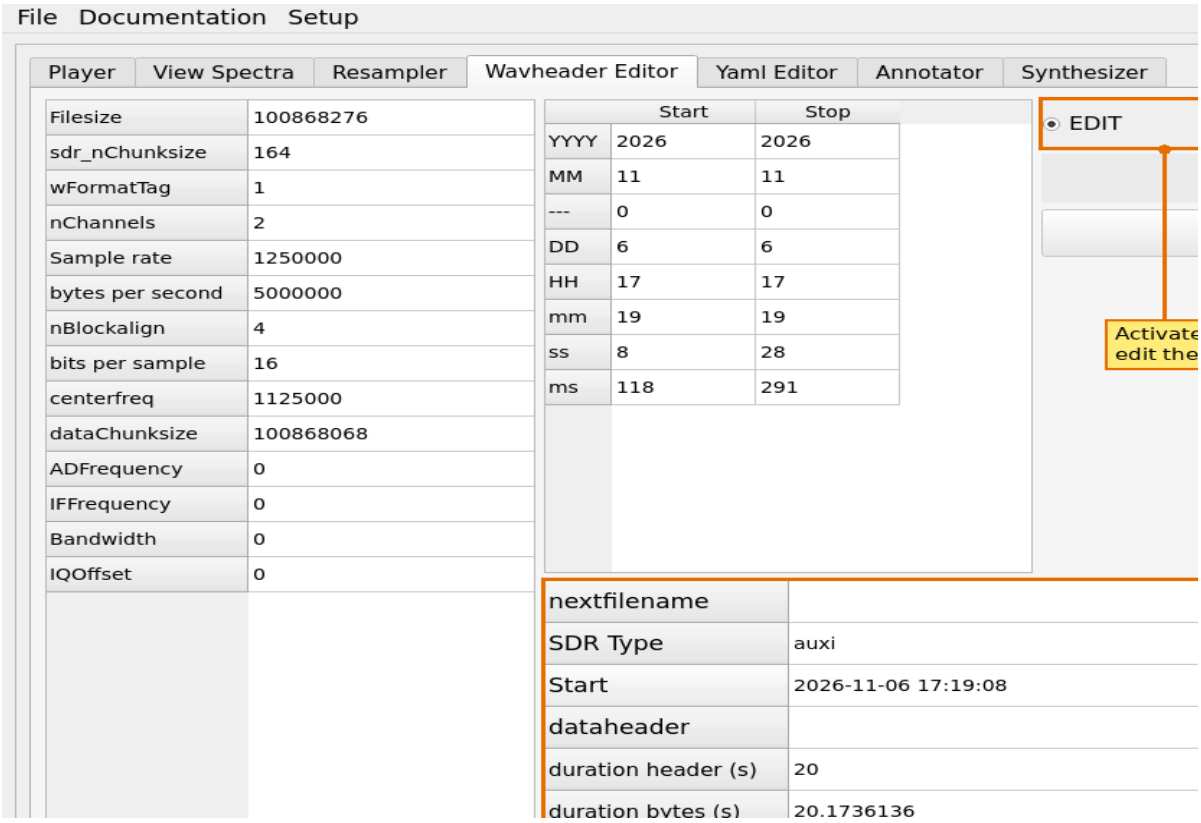


Fig. 1-29: General view of the wav-header editor

The information is displayed in 3 tables. On the left you will find data such as sampling rate, file size, sample size etc. Table 2 at the top right shows the start and end time of the recording in UTC. The third table at the bottom right contains only one really important field: 'nextfilename', i.e. the name of the next file in a series if more than 2 GB were recorded in one go. The fields from 'start' onwards are for information purposes only and are not saved in the header. The 'duration' information in seconds and bytes can be interesting here if you want to edit the yaml files. Be extremely careful with the field SDR-type (here ‘auxi’). Changing this field may cause unexpected results and even prevent readability of the file.

There are 3 important manipulation elements:

(1) ‘EDIT’ allows the entries to be changed when activated, should this be necessary.
(2) ‘Overwrite Header’ writes the edited header into the file (shortcut Alt-H). It is only active while ‘EDIT’ is activated.
(3) Insert Header' is still inactive and, once activated, allows you to convert a 'dat' file recorded with STEMLAB and RFCorder into wav-format.

You now have the following options:

1.8.1 Changing the header of a file:

To do this, a file must be loaded and the 'EDIT' button activated. After changing the header entries, you can click on the button 'Overwrite Header' (or press Alt-H). You will be asked if you really want to do this and if you confirm, the original header will be overwritten. As already said, you should only do this if you know exactly what you are doing. Hint: As the original wav format does not support file sizes > 2GB, the program does not accept entries > 2147483648 in the field 'Filesize' In larger files this entry will thus be invalid by default. The 'dataChunksize' field must have a value that is by 208 smaller, i.e. a maximum of 2147483440. If the difference is not 208, the SDR software SDRUno may abort any playback attempt.

1.8.2 Converting 'dat' file to SDR-Uno format:

This function is only available if the file type 'Raw IQ' (*.dat, 'raw') is selected when opening a file, see Fig. 1-30.

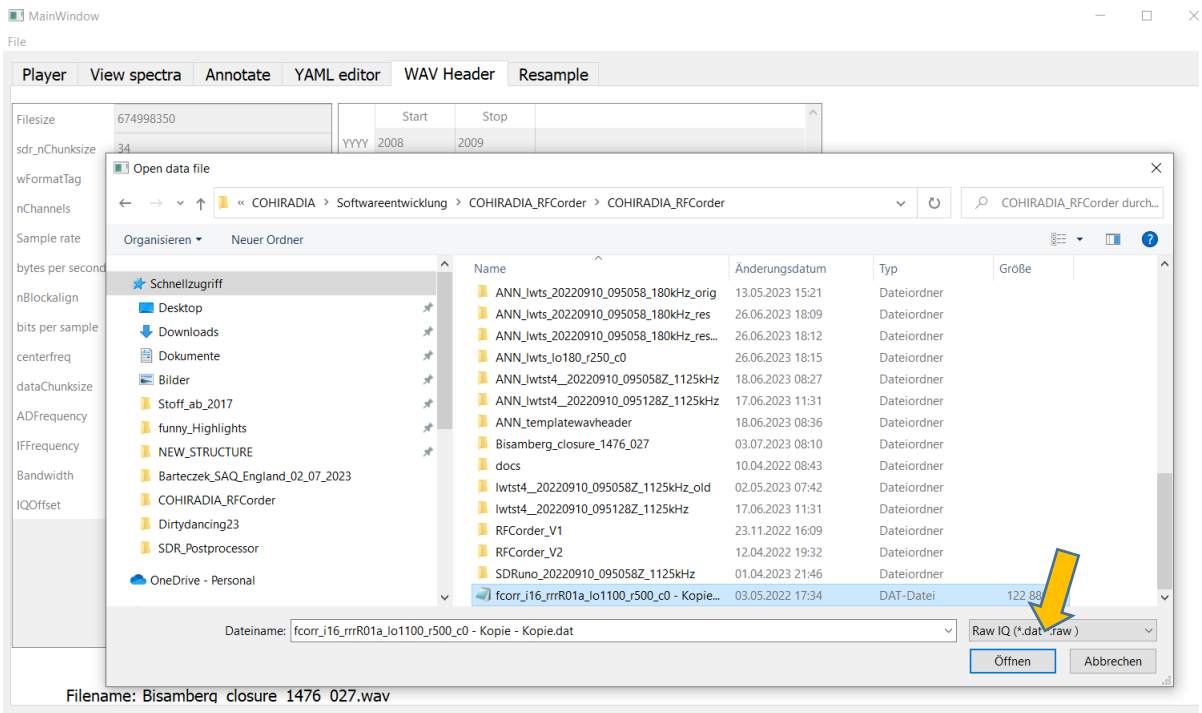


Fig. 1-30: Selection of the file format ‚dat‘ or ‚raw‘, respectively, after having opened a file.

The program automatically creates a basic wav header and then asks for the three parameters 'Center Frequency', 'Bandwidth' and 'bits per sample' via pop-up menus (see Fig. 1-31). 'Center frequency'

corresponds to the local oscillator frequency of the SDR in kHz (band center frequency, coded with '_LO####' in the previous STEMLAB files) and can be entered within certain limits (0 and 60000). This entry determines whether other programs such as SDRUno display the frequency axis correctly and whether STEMLAB plays back at the correct frequency. The 'Bandwidth' menu corresponds to the sampling rate and provides a drop-down list of the sampling rates permitted by STEMLAB. The default setting is 1250kHz for typical MW recordings.

Finally, 'bits per sample' is queried. Here you can select the values '8', '16', '24' and '32' via dropdown. For COHIRADIA, ONLY the preset '16' should be used, so simply confirm with OK.

CAUTION: You can also enter numbers manually in the 'Bandwidth' and 'bits per sample' menus instead of selecting from the drop-down list, but this is not recommended. Manual entries may result in incorrect wav headers in the event of invalid entries (e.g. 1536 kS/s). This is not yet automatically prevented in the current version, so please ALWAYS select from the drop-down lists.

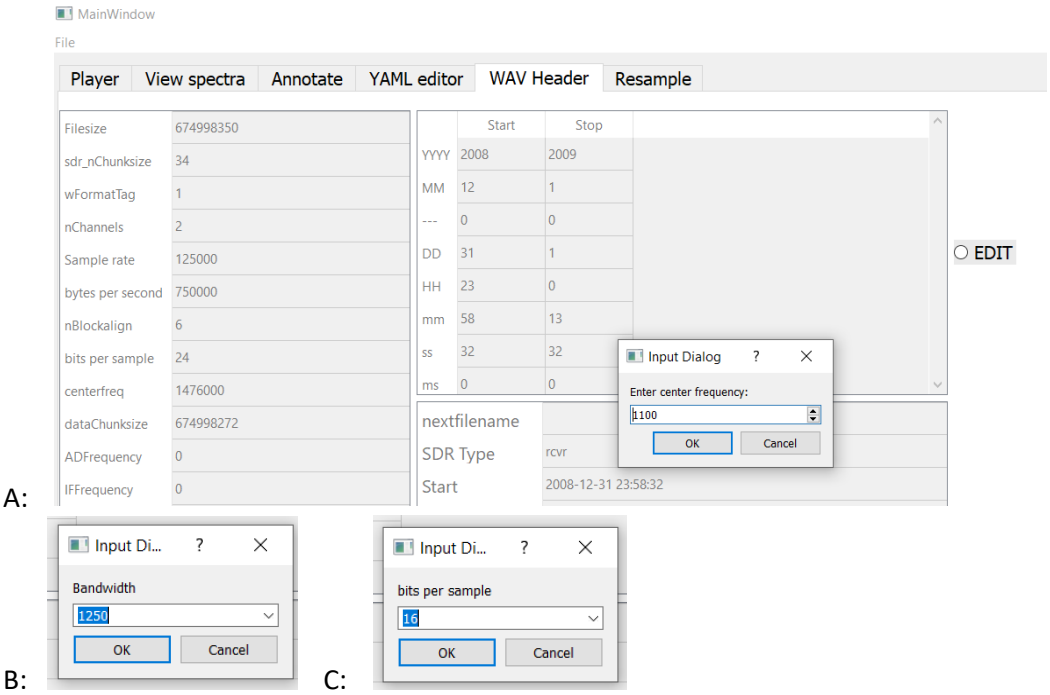


Fig. 1-31: Query for the parameters ‚center frequency‘,‘Bandwidth‘ and ‚bits per sample‘

After the template header has been created, it can be edited via the tables. As shown in Fig. 1-32 in orange, you must now edit the start and stop times, as the automatically generated values are just extracted from the file modification times on the PC and do not correspond to the true recording times:

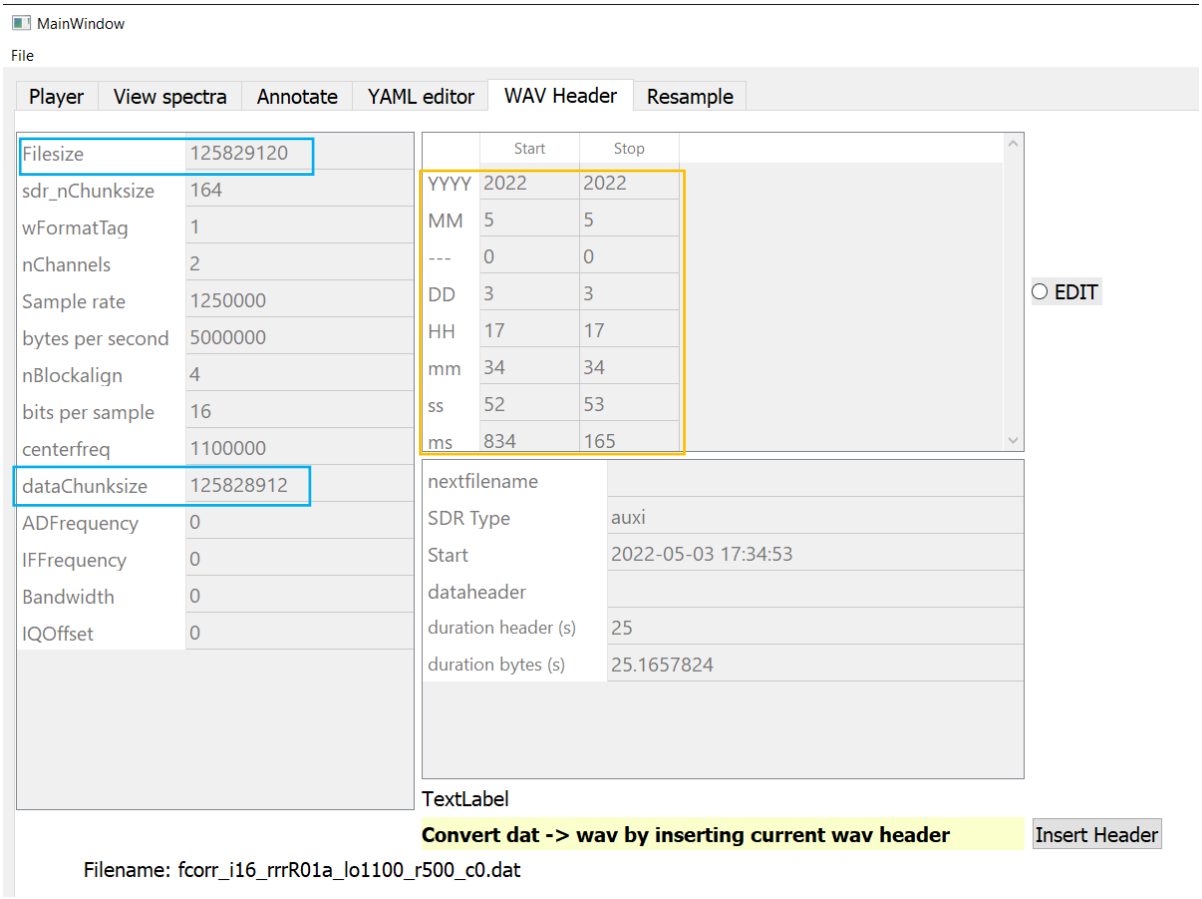


Fig. 1-32: Fields to be edited when converting dat → wav.

The start and end times of the recording must be entered in this time table, in a self-explanatory manner with date (YYYY, MM, DD) and UTC time (HH, mm, ss). The 'ms' field can be ignored. The 3rd line, which is marked with '-', has no meaning, it is filled with meaningless numbers in the SDR-Uno header (reason unknown).

The two blue marked entries 'Filesize' and 'DataChunksize' are automatically entered based on the file size. However, if the 2 GB limit is exceeded (which can easily be the case with STEMLAB recordings made with older versions of COHIWIzard and RFCorder), you may receive an error message later. So please check if values > 2147483648 appear there. In this case, simply enter the maximum for Filesize (i.e. 2147483648) and enter 2147483440 in DataChunksize. Programs such as SDRUno will still play correctly, but SDR# will only play the first 2GB.

To summarize the fields that may need to be checked:

- Filesize in bytes
- dataChunksize must ALWAYS be Filesize - 208
- Sample rate is what is encoded in the RFCorder as '_r####', e.g. '_r1250', but in S/s, i.e. x1000. '_r1250' must therefore be '1250000'.
- Bytes per second: Must ALWAYS be 4 x sample rate, i.e. '_r1250' must be '6000000'
- centerfreq is the LO frequency in Hz, which is coded in the RFCorder as '_lo####', e.g. '_r1100'. For '_r1100' this must therefore be '1100000'

Once you have made these entries, all you have to do is click on the 'Insert Header' button and that's it. The converted file is given a new name according to the SDRUno convention, namely:

[stem of the old filename (without _LO, _r, _c)] + DATE_STRING + TIME_STRING + FREQUENCY_STRING + '.wav'

Example:

‚Samplefile_LO1200_r1250_c0.dat' will be renamed to
‚Samplefile_20220503_173453_1200kHz.wav,

in case the entered starting time and date have been set to 17:34:53 and 2022-05-03, respectively, and the LO frequency is 1200 kHz.

Now programs like SDRUno should play back the resulting file correctly.

1.9 TAB ‚YAML EDITOR‘

To automatically generate a yaml file after annotation, the general part must be filled with the individual station data of the recorder. The table in this tab is used for this purpose, see Fig. 1-33.

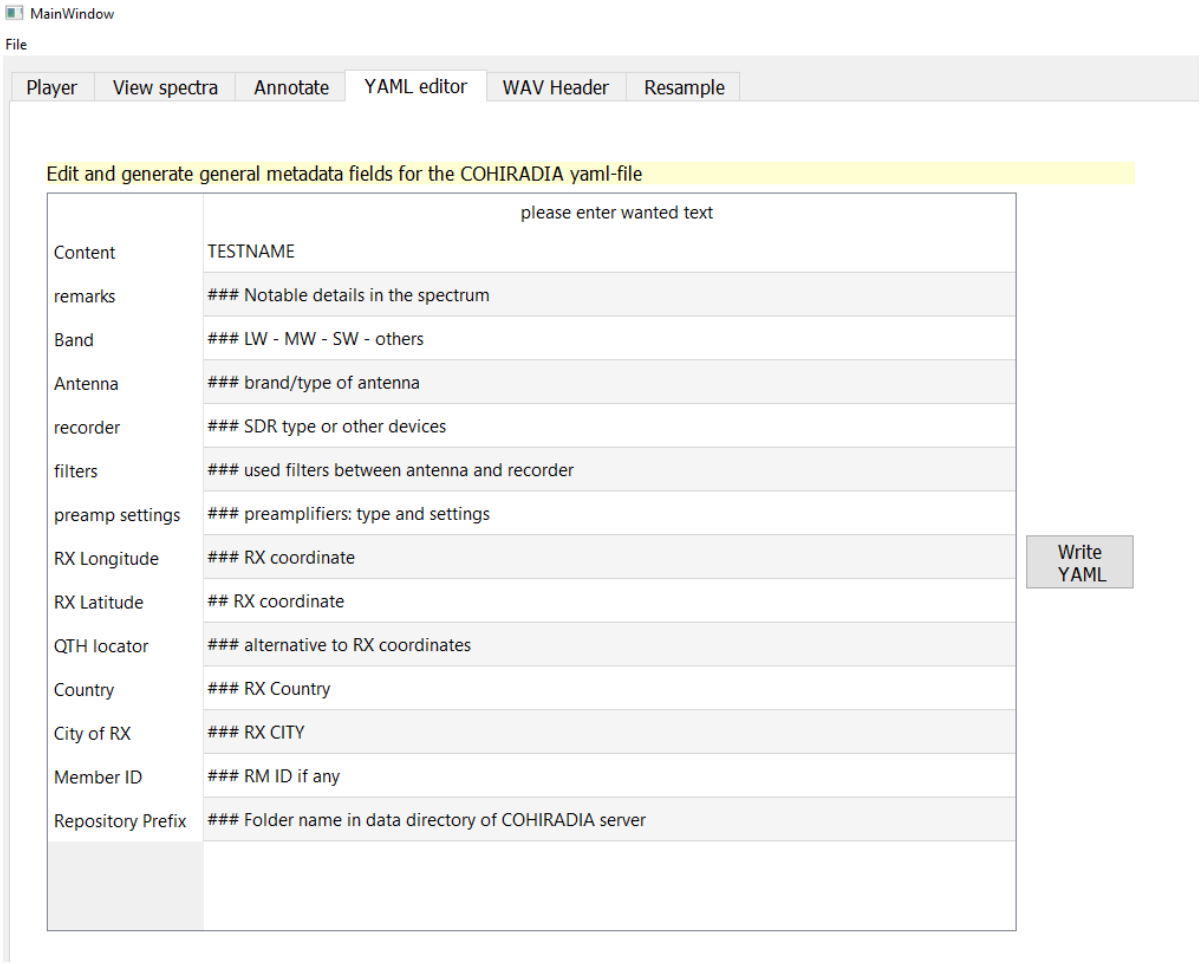


Fig. 1-33: Table for entering the station data and button for automatically creating the yaml header.

The fields should be self-explanatory, simply insert the corresponding texts and finally press 'Write YAML'. The finished YAML is called COHI_YAML_FINAL.yaml and is saved in the directory COHI_root/ANN_[filename]/.

If you annotate a new recording, you do not necessarily have to regenerate this information if the station data is the same or similar. You can copy the two files

COHI_root/ANN_[Alter_Filename]/cohiradia_metadata_header.yaml

and

COHI_root/ANN_[Alter_Filename]/cohiradia_metadata_tailer.yaml

into the new annotation directory

COHI_root/ANN_[New_Filename] .

When the COHIWizard is restarted and the file to be annotated is reopened, the information is then loaded automatically.

Programmer's note: New versions will automatically load a copy of the last station profile used, but this is currently still a work in progress.

1.10 Tab Resample

This tab is still under development and thus some planned features are not yet enabled, though pre-configured in the GUI (e.g. cutting tool, frequency correction for originally analogue VCR recordings) .

IMPORTANT ! Absolute requirement: The freely available command line program 'sox' must also be installed on the PC. Sox can be downloaded for free at:

https://sourceforge.net/projects/sox/files/sox/14.4.2/

It is best to install sox in the installation directory of COHIWizard. Alternatively, the system path must be set to the installation directory of sox.

Fig. 1-34 shows the interface of the resampler after loading a file with 'open File'. The spectrum is displayed in the plot window, the current center frequency and the bandwidth are marked by the orange thread and the red rectangle. Outlined in blue you can see the fields for setting important target parameters like the new sampling rate, the new center frequency and optional setting of the gain.

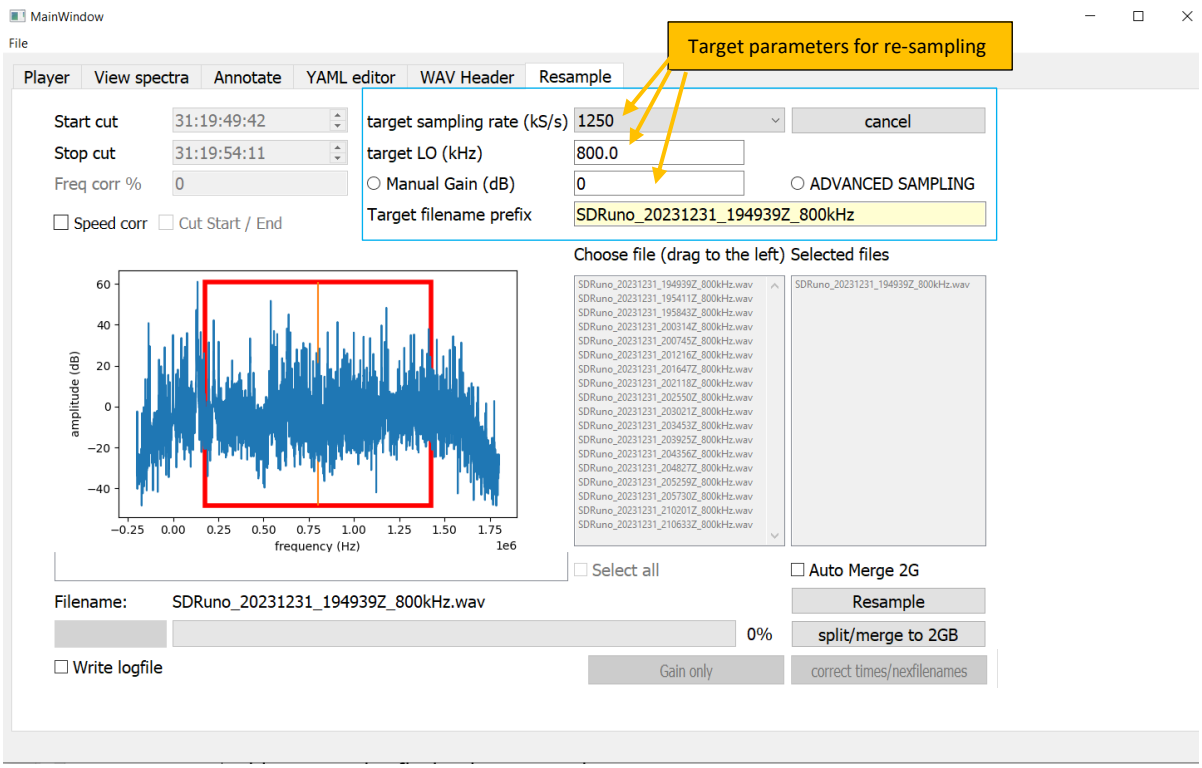


Fig. 1-34: GUI of the resampler

1.10.1 Standard application:

The most common application is probably to recode a recording if its sampling rate does not correspond to one of the values allowed for STEMLAB (e.g. RSP1a, long wave: SR = 150kHz). The spinbox 'target sampling rate' should be set to the next higher available rate, e.g. 250 kHz in case of an original bandwidth of 150kHz. The center frequency (= target LO) should be left as it is. This increases the size of the file, but is necessary for playing back the whole band. If you then press the button

'Resample' in the bottom right corner, the conversion procedure starts and creates the new file, which gets a new name containing the string '_resamp'. The conversion requires several steps and takes some time, but of course it has to be done only once. The progress of the procedure is displayed with a progress bar below the spectrum window. After successful conversion the new file can be loaded and played back.
During the conversion, the file is automatically converted to 32bit COMPLEX, i.e. 16bit per sample. This means that originals with 24bit (e.g. many recordings made with PERSEUS) or 32bit can also be converted to COHIRADIA format.

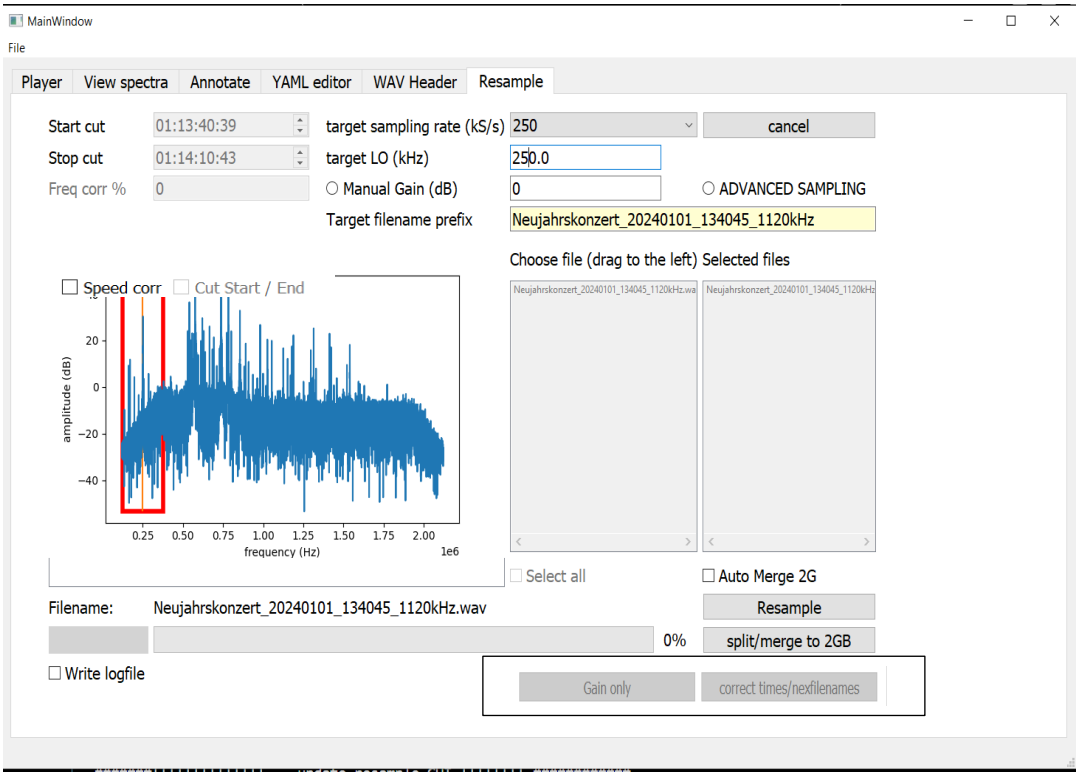


Fig. 1-35: Cutting the LW-Band out of a 2MHz broad recording made with a PERSEUS SDR.

1.10.2 Advanced Applications:

For 'advanced use' there is also the option of selecting a different sampling rate and even shifting the center frequency if necessary. This is usually not required by the standard user, but if you want e.g. to cut out only the LW band from a very broad spectrum ranging 0 to 2000 kHz (typical e.g. for PERSEUS-SDRs), this function is useful. This case is illustrated in Fig. 1-35. With 'target sampling rate' you can set the target sampling rate for playback on the STEMLAB. For LW, 250kHz makes sense, as this then covers more than the entire band. The LO frequency selected here is 250 kHz, which results in a target band of 250 +/- 125 kHz, i.e. 125 - 375 kHz. If, alternatively, you only want to cut out the medium wave band, it is advisable to set the sampling rate to 1250 kS/s and the target LO to 1100 kHz. If you now click on 'Resample', a two-stage procedure begins in which the spectrum is first shifted and then resampled. The procedures may differ depending on whether the IQ data is 16, 32 or 24 bit. You now have the following options:

A) Resampling a single file

B) Resampling an entire list of related files

C) Splitting an existing file into 2GB tranches without resampling

1.10.2.1 Case A: Resampling of a single file:

Resampling is carried out by simply clicking on 'Resample' after loading a file. The resulting file is copied to a new directory 'out' (subdirectory of the directory where the source file is located) and has a file size that has been changed according to the ratio between the source and target sampling rate. The file is named [old filename]_"resamp"_[new date]_[new time]_[LO frequency] "kHz.wav".

In the specific case in Fig. 1-35, the name of the target file is: 'New Year's Concert_20240101_134045_1120kHz_resamp_20240101_134045_250kHz.wav'.

Everything is fine if the output file is not more than 2GB in size and the target sampling rate is not greater than the output rate (downsampling). In many cases, however, this is not the case. If you want to ensure that the target files do not exceed 2GB, you should therefore check the 'Auto Merge 2G' option. Then 2GB files will always be created and the filenames will take the form [Target filename prefix]_[Number]_[date]_[time]_[LO frequency]"kHz.wav". The intermediate files with prefix '_resamp_' are deleted at the end.

[Target filename prefix] is the string entered in the "Target filename prefix" field. For the prefix 'my_new_file' entered in Fig. 1-36, the only possible target file is 'my_new_file_1_20240101_134045_250kHz.wav'. If the source file had a significantly greater length (e.g. 20GB), then a further 2GB file 'my_new_file_2_20240101_141632_250kHz.wav' would be created.

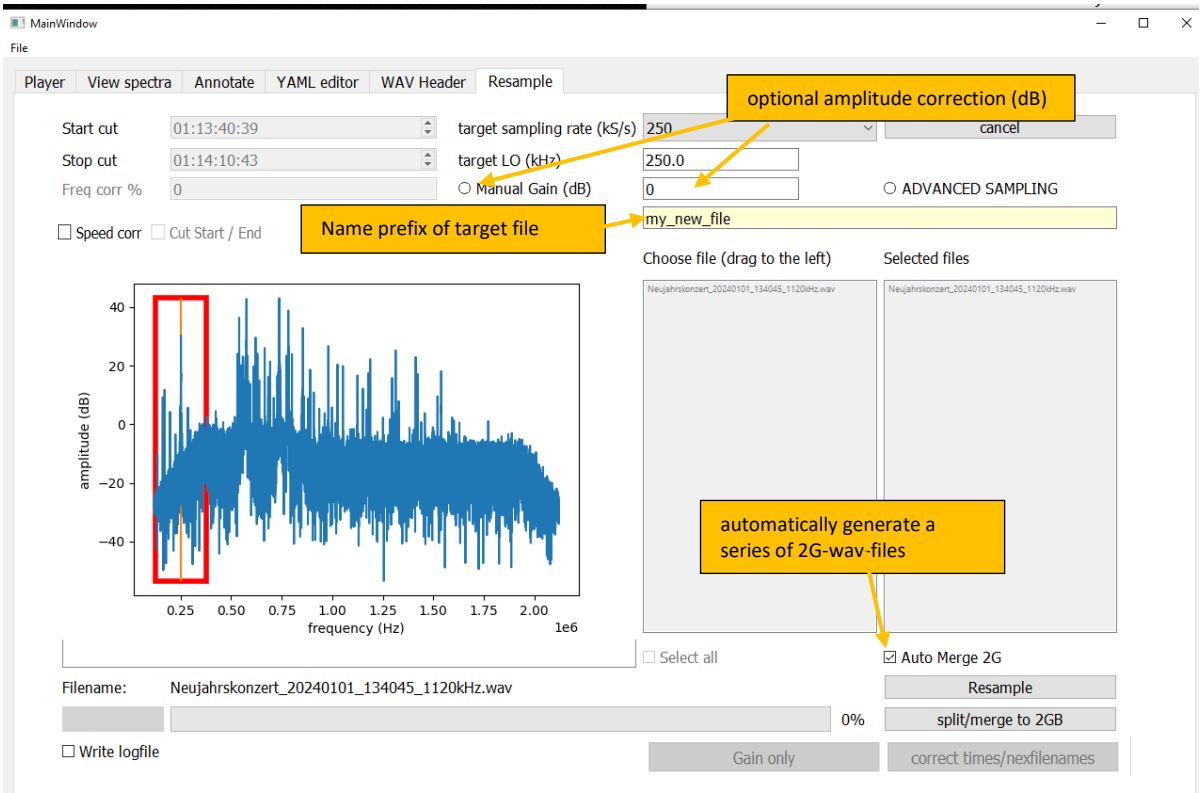


Fig. 1-36: Activation of the automatic segmentation in 2-GB files and setting the prefix of the target filename.

If you have forgotten to check 'Auto Merge 2G', you have to reload the 'resamp' files and then combine them with 'split/Merget o 2GB'. However, the '_resamp_' intermediate files are then not deleted in the process and you have to remove them manually, if wanted.

The tool also gives you the option of correcting the amplitude of the RF signal if it is too small. Such a small amplitude occurs more often than it may be expected and then any playback in the Player may results to unsatisfactorily low volumes. In the newer software versions you can then use the volume control to increase the volume, but this option is not yet available in the oldest RFCorder versions. It is therefore advisable to check and, if necessary, correct the amplitude. This can be done by clicking on the radio button 'Manual Gain (dB)' (see Fig. 1-36). The value must be entered in dB in the adjacent text field. To check whether the amplitude is correct, you should switch to the 'View spectra' tab and tick the 'plot raw data' option, see Fig. 1-37. The correct amplitude AFTER applying gain is always displayed there. The signal should ideally be between 0.1V - 0.7V, but in any case remain below 1 V, i.e. better with a safety margin, otherwise bad intermodulation distortions will occur. If, on the other hand, the signal is mostly below 0.001V, you should definitely amplify the signal.

ATTENTION: The Gain function is only active when resampling, it does not take effect if you only execute 'Split/Merget o 2GB'.



Fig. 1-37: Checking for correct signal amplitude e.g. when setting the gain before resampling. The signal values need always to remain below 1V to prevent clipping. It is recommendable to leave some margin for reducing intermodulation artifacts.

1.10.2.2 Case B: Resampling of a list of associated files:

Often you have made a longer recording with an SDR, which then typically consists of several wav files, each 2GB in size. As one example SDRUno creates such file series, in which the header of each file refers to the next one, so that the software can play back the entire series without interruption. This philosophy has also been adopted for the player of the COHIWizard.

If you want to resample such a whole series, it is advisable to copy all the related files into a separate directory and open any one of them with 'File open'. All the wav files in the directory are then displayed in the 'Choose files' list, although they are initially grayed out. If you now activate the 'ADVANCED SAMPLING' option, you can drag and drop individual files into the 'Selected Files' field, see Fig. 1-38 A.

If you want to resample the entire list, you can also check 'Select all'. If you have selected a file by mistake, you can move it back again by dragging with the mouse, or you can deselect the entire list by unchecking 'Select all', see Fig. 1-38 B.

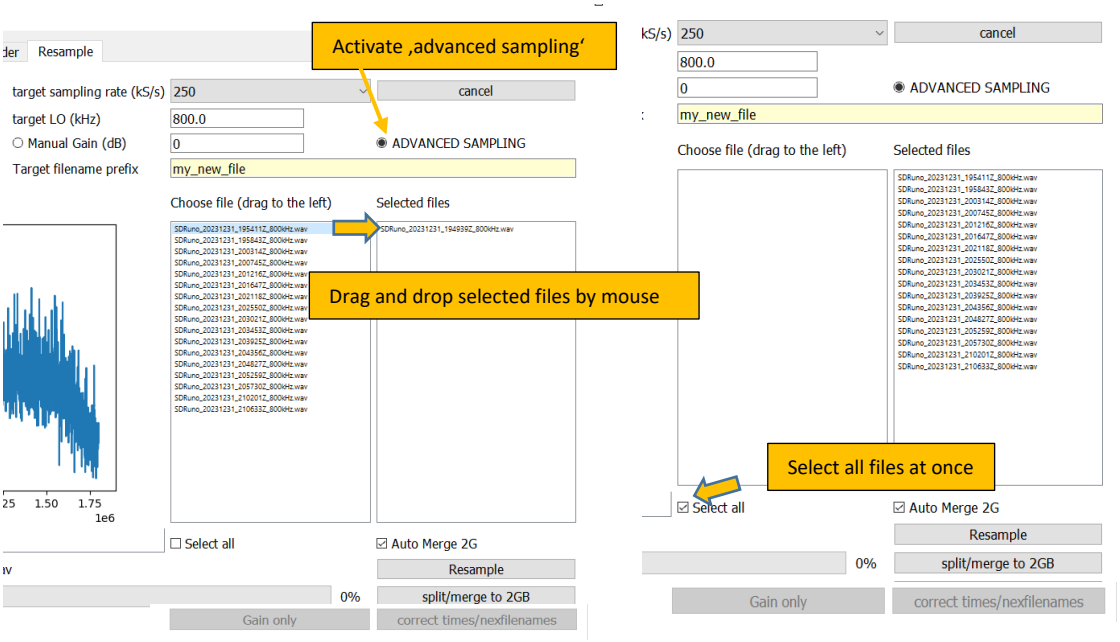


Fig. 1-38A: Selection of files to be resampled by drag and drop by the mouse.

B: Select/unselect all files of the list at once by checking/unchecking the corresponding checkbox, respectively.

ATTENTION: the files in the 'Selected Files' list must appear in the correct chronological order, as the target files are of course also created in this order. If this is not observed, not only will the result not be correct, there may also be unforeseen complications when writing the new wav-headers. The system does not currently check excessively whether the listed files fit together correctly. Currently, the system only checks whether there is a time difference of no more than 300s between the start time of a file and the stop time of the previous file. If this is not the case, the program aborts the process, but only during the resampling process.

Special feature for treating 'holes' in recording series: The tolerance of 300s can be used if, for some reason, you have produced a small time delay between two consecutive files during recording. This could be because you had to stop and restart the recording due to an error, or because the recording software itself was too slow when switching to the next file (this can happen with large bandwidths and slow PC hardware). In this case, the time display of the target recording would be incorrect because segments are missing. In this case, COHIWizard inserts an 'empty signal' into the target series at the relevant points (the signal then just consists of silence) in order to ensure a target file with the correct duration and correct time stamps. In such cases, gaps of up to 300s are tolerated and replaced by silent sequences.

1.10.2.3 Case C: Splitting of an existing file into 2GB segments without resampling:

If, for example, you have created a file with a size of more than 2GB with the old RFCorder or another tool and you want to split it into 2BG tranches with correct wav headers, you can proceed as follows:

(1) If it is a dat file, it must first be provided with a wav header, see section 1.8.2.
(2) If it is a wav file, simply click the 'Split/Merget to 2GB' button and a corresponding file series will be created in the 'out' folder.

However, as the name of the button suggests, this function can also be used to copy series of files with more or less than 2GB into a series with 2GB per file. All time stamps in the wav headers are automatically converted and entered correctly.

ATTENTION: The ‘Gain’ function is only executed when resampling, it does not take effect if you only execute 'Split/Merge to 2GB'.

ATTENTION: You should ensure that you have sufficient disk space in the target directory, as resampling can produce relatively large result files. Some temporary files are also created. The program does check the available space at some points and should issue warning messages if there is too little available, but the reliability of these displays has not yet been tested excessively.

1.10.2.4 Cutting

It is possible to trim the start of a recording, i.e. cut out the part before a certain time set in the field ‘Start cut’. This function is only activated if you have checked ‘ADVANCED SAMPLING’ and ‘Cut start/end simultaneously (see fig. 39). Currently trimming of the end is not yet possible. Although this function is already provided in the GUI (Stop cut), they are still grayed out. it will be activated in future releases.

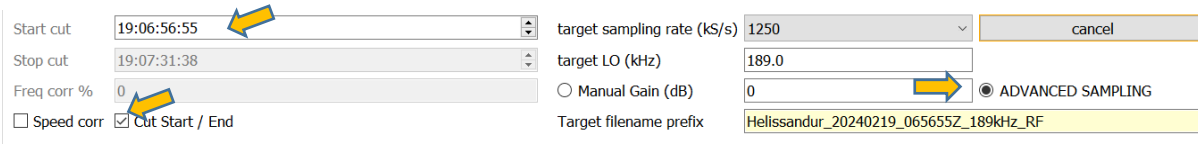


Fig. 1-39: Special functions for trimming and speed correction.

1.10.2.5 Speed corrections

This is a very special option which may be rarely used. It is provided for cases where the original recordings have been made in an analogue way, e.g. by a videorecorder. In such cases the speed of recording and playback differ by a few percent. Then the carrier frequencies are not located correctly and a tape speed correction is necessary. This is possible by checking ‘speed corr’ and specifying the %-value in ‘Freq corr %’ (see fig. 39).

1.11 TAB SYNTHESIZER

The synthesizer allows to create RF bands with custom-defined carrier frequencies which can be modulated in AM with custom-defined audio signals. There exist two basic modes, a slow one and a fast one.

The slow mode is a generic implementation under Python. The fast mode is accessing *ffmpeg* which operates by a factor of up to 5 faster than the original one. On the downside this option requires double as much temporary diskspace as the classical (slow) mode, because some intermediate files with the size of the full recording must be stored together with the main file.

Warning: So far the fast mode has a somewhat worse noise floor than the slow mode, so that it is considered as 'beta'-version. Consequently it is not the default mode but must be activated explicitly by clicking the corresponding radio button **'FAST mode (beta)'**.

1.11.1 Prerequisites:

1. Third-party software *ffmpeg* to be installed on your computer
2. Audio source files in form of standard *.wav or *.mp3 format

Ad 1 : In the Windows exe-Version of COHIWizard V2.x *ffmpeg* is automatically installed in the folder **ffmpeg-master-latest-win64-gpl-shared.** Should you use the source codes from GITHub and call the main program from the Python prompt and *ffmpeg* is not installed on your computer, then you are asked during the startup whether you would like to install it. Clicking 'yes' will install *ffmpeg* in the **ffmpeg-master-latest-win64-gpl-shared** folder of the COHIWizard file system. Currently there is no progress bar available which shows the progress during installation, you can only see a few status messages on the terminal window in the background. So please just stay patient until the GUI becomes responsive again, this can take up to a few minutes.

On LINUX systems (e.g., Debian) *ffmpeg* is frequently already pre-installed, so that you do not have to worry about it. Should this not be the case, you have to install it manually.

Ad 2: It is recommendable to organize the audio files you would like to use in only few dedicated folders on your computer or external hard disk so that you need not choose the source directories frequently during project composition.

1.11.2 Basic operation

On the toolbar you can find a tab 'Synthesizer'. Clicking on it will present the GUI depicted in fig 1-40 .

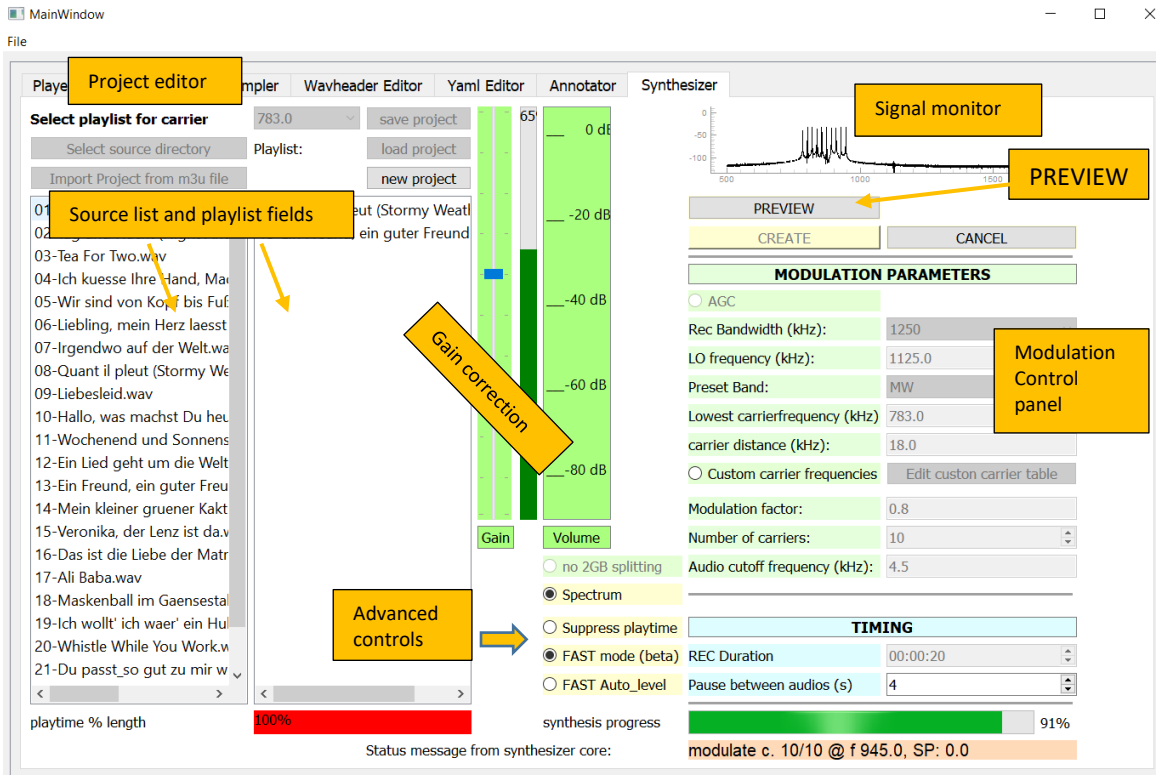


Fig. 1-40: Synthesizer main panel.

1.11.2.1 Setting the modulation parameters

To compose a new RF band first you should define the basic parameters in the field 'modulation parameters' panel as illustrated in fig. 1-41.

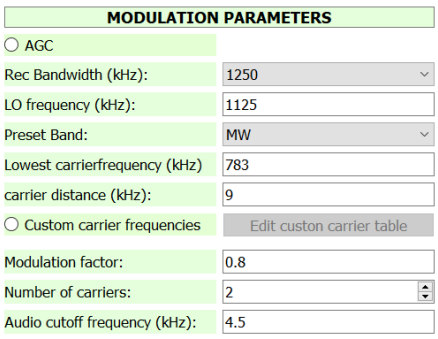


Fig. 1-41: Default Configuration menu for the modulation parameters

The radiobutton **'AGC'** is not yet functional in v2.2.0, it is reserved for optimizing the signal envelope for the dynamic range in future releases.

Rec Bandwidth defines the width of the band to be synthesized (BW). You can select pre-defined values which are admissible for the STEMLAB125-14. For MW e.g., a value of 1250 kHz is recommendable. You can also select pre-set recommended bandwidth values for the most frequently used AM-bands from the dropdown menu **'Preset Band'** (MW, LW, 49m, …).

LO frequency refers to the center frequency of the band and can be set to any reasonable value in the allowed frequency range of the STEMLAB 125-14 (i.e., from BW/2 up to 60 MHz – BW/2). The default value 1125 is a typical one for MW.

'Lowest carrier frequency' allows to define the lowest frequency used in the band.

'Modulation factor' is the AM modulation factor and is set to 0.8 by default. This is usually a good value, but you may have reasons for changing it to other values. Values above 1 are not recommendable.

The carrier frequencies can be defined either with a fix grid of N carriers with equidistant spacing by or in a custom-defined individual table.

(1) Definition with fixed grid: This is the easier way and mostly sufficient. Just select N in the field **'Number of carriers'** and D being in the field '**carrier distance'**. The default starting values are 2 and 9kHz, respectively, but you can set any reasonable value. The carrier distance, however, needs to be at least twice the **Audio cutoff frequency** as defined in the last field. By default the latter is set to 4.5 kHz (for MW). If the number of carriers is not excessively high (e.g. <=50) and the choice of the exact frequencies is not crucial, then a carrier distance of e.g. 18 kHz may be preferable because it allows for better channel separation, especially on very old radios with poor selectivity.

(2) Definition with a custom table. To this end activate the radio button **'Custom carrier frequencies'** as shown in fig. 1-42. Now click on '**Edit custom carrier table'**. A table widget opens and you can there fill in as many arbitrary frequencies as are allowed when considering the audio cutoff and the overall 'Rec Bandwidth'. Clicking 'OK' will close the table and save the settings.

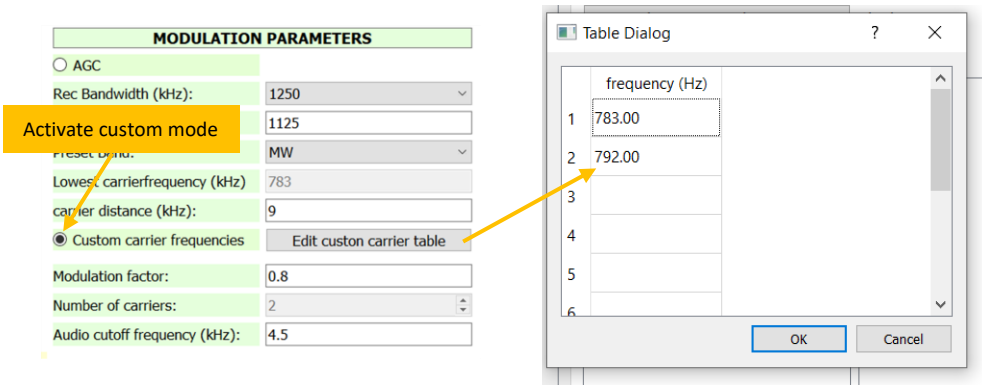


Fig. 1-42: Activation of '**custom carrier frequencies'** and clicking on **'edit custom carrier table'** let's pop up a table dialog for entering individual frequencies

1.11.2.2 Assembly of the playlists

After having set all these parameters appropriately, you can click on the combo-box in the top left quadrant of the GUI and the list of all set frequencies will drop down (see fig. 1-43). Clicking on one of them will set this frequency as the current one to be modulated with audio contents.

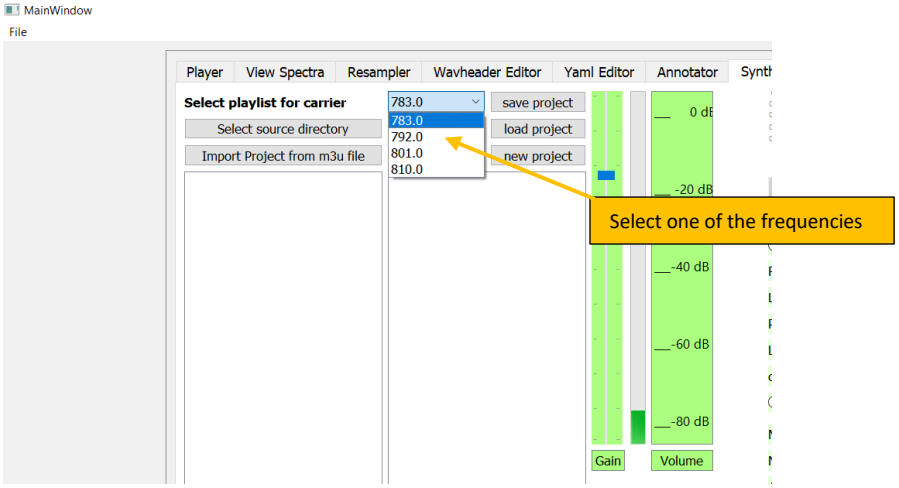


Fig. 1-43: Selection of one of the listed frequencies for modulation with an audio playlist

Once having defined all the described settings, you can start to compose the playlists for the individual carriers. To this end click on **'Select source directory'** and select the directory where your audio files are stored, see fig. 1-44.

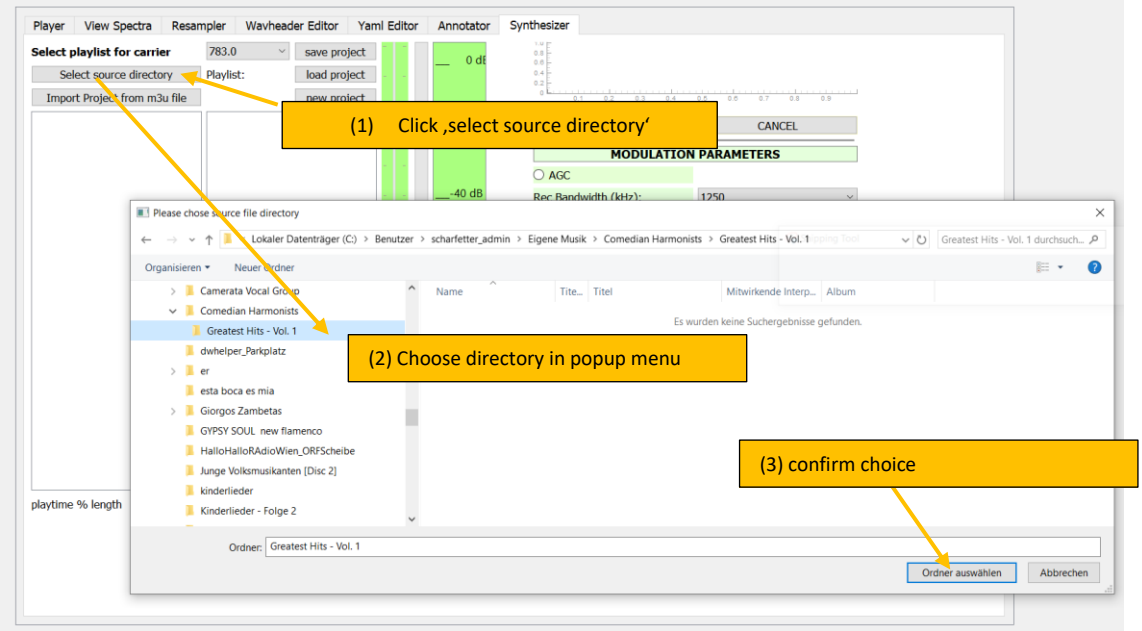


Fig. 1-44: Selection of one of the listed frequencies for modulation with an audio playlist

Once done this, the content of the directory will show up in the left list widget and you can drag titles from it to the right list widget, which will then constitute the playlist. You can also change the order of titles in the playlist just by clicking and dragging the list elements up/down. If you would like to remove a list element, right click on it and chose 'delete' from the context menu. The progress bar labelled **'playtime % length'** indicates how much of the overall playtime (set in the field '**REC duration'** in the 'TIMING' panel) is already covered by the selected titles. Should the list run out of time the progress bar turns red. In this case, titles after reaching the overall playtime will be cut off. In order to prevent this, either shorten the list or increase the 'REC duration' (format hh:mm:ss). This is illustrated in fig. 1-45.

By default, there is a pause of 4s between subsequent titles so as to keep some 'silence' between them. You can change this value in the field **'Pause between audios (s)'.**

Should not all titles for one carrier be located in the same directory, you can select also other ones and drag titles from there. After compiling one carrier, like in fig. 1-46 you can select the next one according to fig. 1-43 and assign the corresponding audio titles. Finally, you should have assigned playlists to all carriers.

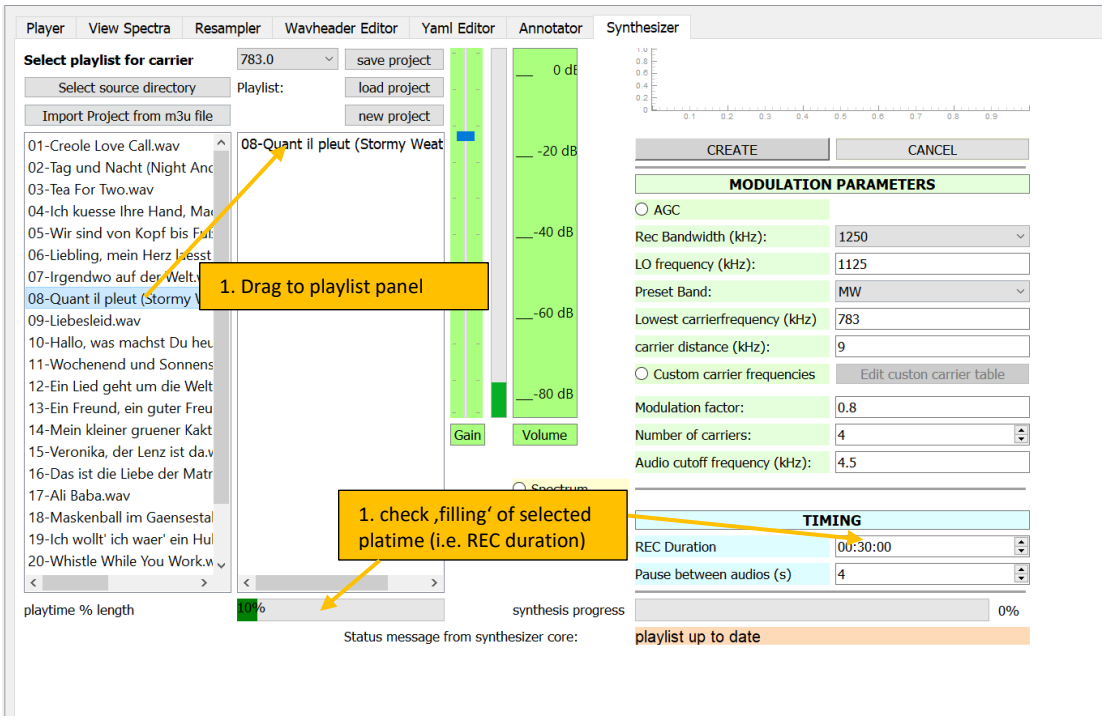


Fig. 1-45: Drag titles from the source list to the playlist for selected carrier, here 783 kHz. Observe the percentage of the playtime being already covered as displayed in the progress bar '**playtime % length**'. If playtime is exceeded, increase the value in 'REC duration'.

1.11.2.3 Save/load/clear project

You should save the project from time to time by clicking on the **'save project'** button next to the carrier dropdown combobox, as shown, in fig. 1-46.

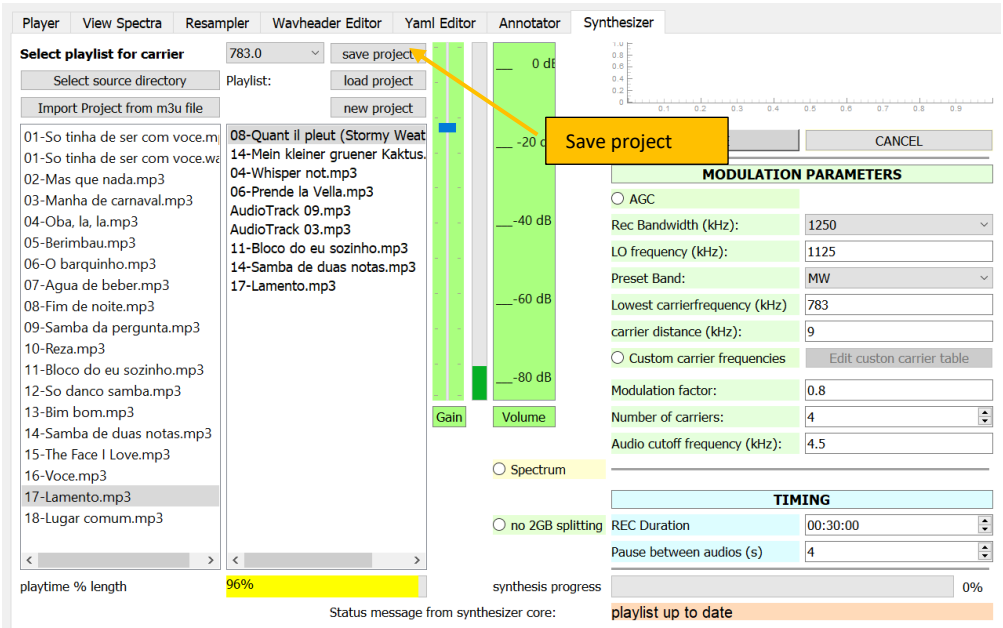


Fig. 1-46: Example of one 96% full playlist for one carrier.

You can interrupt working in one project at any time, close the COHIWizard and re-start it later. You can then load the project again pressing **'load project'** and continue editing.

You can also change the settings of the modulation parameters any time but be aware that when removing carriers (reducing 'Number of carriers'), then you lose the respective playlists. Warnings will be issued when you try such changes. Thus it is recommendable to define all modulation parameters correctly at the beginning of a project in order to avoid later confusion. This is especially true for the number of carriers, the carrier distance and the audio bandwidth.

Should you wish to create a new project and clear everything which is currently loaded, then press the button **'new project'**.

1.11.2.4 Set the path for the synthetized SDR-wav files

Synthesized files will be stored in the path specified in the field **'Edit recording path'** in the **player tab** (there is no such a choice in the synthesizer tab). Thus set the path appropriately.

1.11.2.5 Preview SDR wav file

As the modulation process may last long time, it is recommendable to first produce a short 'test' file and check if it fulfils the expectations. Clicking on the 'PREVIEW' Button starts the assembly of an SDR-Fie with 20s of duration. This is especially interesting when activating the 'FAST' mode with the respective radio button (see fig. 1-40b). In this mode the complex digital modulation procedure is carried out via a special filter chain of *ffmpeg*, which does not allow for generating a full spectrum blockwise, but which builds up the whole full-length recording carrier by carrier. Thus, for seeing the full spectrum one has to wait nearly until the end of the synthesis procedure. Once satisfied with the result, the full synthesis can be started with the button 'CREATE', see next sub-section.

1.11.2.6 Create SDR wav files

Once a project is completely defined, you can create the SDR-wav-file by clicking on the **CREATE**-button above the modulation parameters panel. Then the synthesis starts. Depending on the resources available on your computer this takes a while, so you must be patient during the assembly of the wav-File. The progress is shown in the **progress bar 'synthesis progress'**. After initial steps for signal scaling the signal monitor in the top right corner of the GUI starts showing the raw time signals, from which the signal envelope can be recognized. While this monitor is available both for the classic as well as the new 'FAST' mode, the information displayed is different.

A: Classic mode ('FAST mode (beta)' **is unchecked):**

> The trace should best remain between the red horizontal lines which mark ca. 75% of the dynamic range, see e.g., fig. 1-47. Should the envelope exceed the range +/- 1 clipping and strong intermodulation errors will arise. In that case the gain can be reduced with the 'gain' slider in the middle of the GUI. However, there is an automatic clipping checker built in the code which normally keeps the signal within acceptable bounds. Should the signal be far below the red limits, then the gain can be increased so as to fill the dynamic range appropriately.
>
> WARNING: the new gain takes only effect from the moment on where it has been set, so the signal before that time stays unaffected. From V2.x on there is a preview mode which facilitates clipping control beforehand.

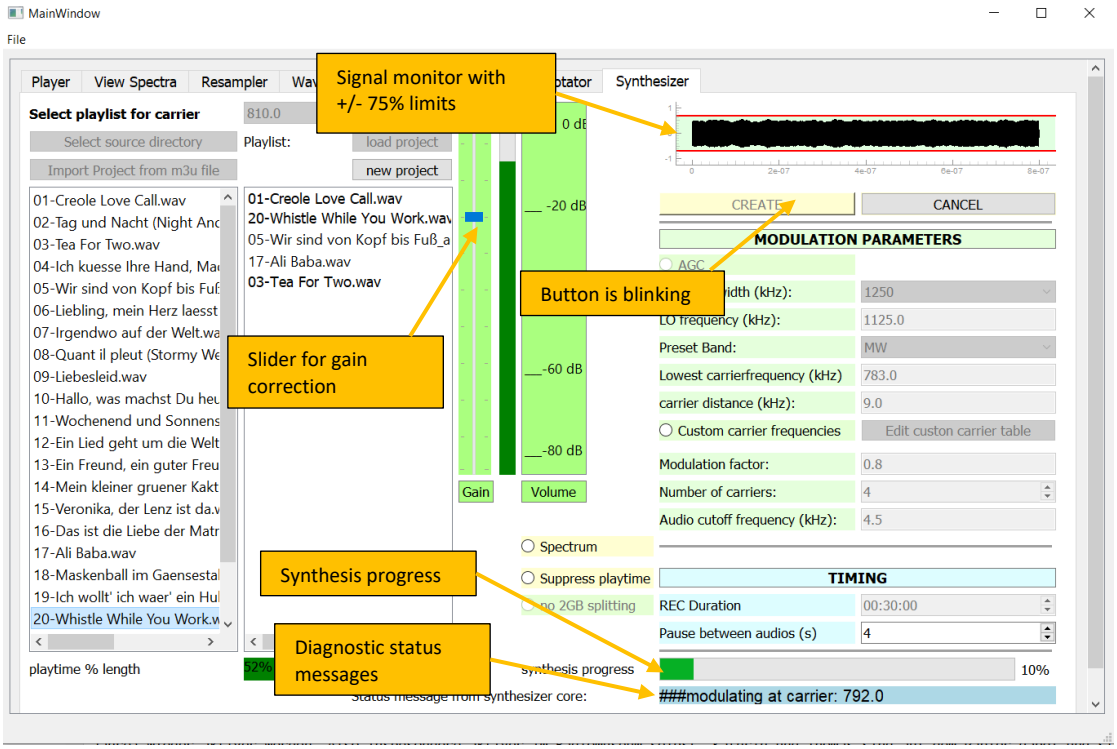


Fig. 1-47: Synthesis phase.

B: FAST mode ('FAST mode (beta)' is checked):

In this mode the amplitude of the monitoring signal is not representative for the final amplitude after completion of the synthesis. As *ffmpeg* is adding one carrier after the other, the signal amplitude grows from very low towards its final value at the end of the synthesis. Potential clipping cannot be recognized during most of the time. Thus it is recommendable to check the signal in the PREVIEW mode. The gain value set at the beginning of one run will be applied during the whole run. Thus, after changing the gain a new run has to be carried out with the new setting. Unlike in the classical slow mode there is no interactive adaptation of the gain during one run before completion. Changes take only effect in the next preview run.

During synthesis, the CREATE button will blink and status messages will appear at the bottom right corner of the GUI, showing some detail information about intermediate synthesis steps.

Should you be interested in the generated spectrum rather than in the raw signal, check the radio button 'Spectrum' below the gain slider item, as shown in fig. 1-48. Then the signal monitor will show the spectrum. However, no clipping information is available in this mode.

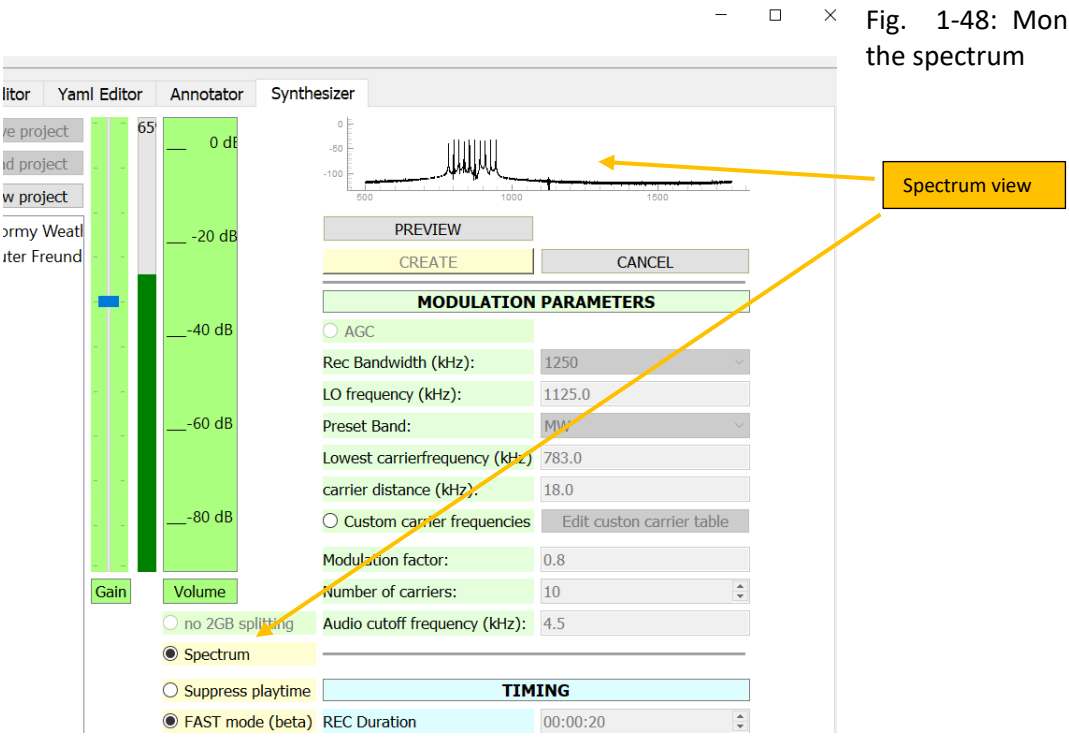


Fig. 1-48: Monitoring of the spectrum

1.11.3 Splitting into 2GB tranches:

A: Classic (slow) mode:

In the classic mode, by default, the synthesizer generates standard SDR-wav files with the appropriate wav headers so that they fully comply with the format of the COHIRADIA recordings. Also, the recording is split into 2 GB tranches in accordance with the usual file size standards used by many SDR software products, like e.g., SDRUno. Should you prefer to generate the recording as a single big file, then check the radio button "no 2GB splitting" below the gain slider item. This mode, however, is not recommended in general, as some SDR software products cannot cope with such files correctly (e.g., SDR#, which terminates playback after 2GB). With COHIWizard you can play back both versions in any case in the player module.

B: FAST mode:

ffmpeg cannot, by default, generate 2GB tranches with the correct wav-headers. Therefore, unlike in the classic mode, the result is a single, typically very large file. As described in 'A' this may be undesirable for compatibility reasons with some SDR software products. In this case you can still split the file by using the 'split/merge 2GB' tool of the resampler, see section 1.10.2.3. In this mode the radio button 'no 2GB splitting' is of course meaningless and just ignored.

If you wish to cancel the synthesis, you can click the **'CANCEL'** button. This will stop the procedure. Files generated so far will not be deleted, hence the SDR-wav files generated up to that moment are valid and can be played back as usual. If you wish them to be deleted you must do so manually or overwrite them in the next synthesis run.

1.11.4 FAST Auto_level

Checking the radio button 'FAST Auto_level' enables *ffmpeg*'s option for automatic gain correction of the concatenated audio files of one carrier so that the audio signal fits reasonably to the allowable dynamic range. Anyway, it is recommended to use only audio files with good quality and well-normalized dynamics.

1.11.5 Automatic generation of playlists via m3u, import of remote audios via URLs

For advanced users there is also a possibility to import playlists from special *.m3u files, which can also be generated automatically by appropriate tools. This option can also handle remote audio files accessible via the WEB via URLs. For this purpose, click the **'Import from m3u file'** button and select the respective file. The playlists will then be built up automatically. The format of the m3u-file is as follows:

#EXTM3U

#PLAYLIST: <name of the playlist>

#EXTPARAM_BW: <channel bandwidth in kHz>

#EXTPARAM_RECLENGTH: <overall playlength in format hh:mm:ss>

#EXTGRP: <carrier_frequency 1>

#EXTINF: -1, <Artist> - <album>

<Path to audio file 1 of carrier 1>

#EXTINF: -1, <Artist> - <album>

<Path to audio file 2 of carrier 1>

……repeat until end of playlist for carrier 1..

#PLAYLIST: <name of the playlist>

#EXTGRP: <carrier_frequency 2>

#EXTINF: -1, <Artist> - <album>

<Path to audio file 1 of carrier 2>

#EXTINF: -1, <Artist> - <album>

<Path to audio file to audio file 2 of carrier 2>

……repeat until end of playlist for carrier 2..

Repeat #PLAYLIST blocks for all remaining carriers

Remark: If the fields

#EXTPARAM_BW: <channel bandwidth in kHz>

#EXTPARAM_RECLENGTH: <overall playlength in format hh:mm:ss>

Are not specified (lacking), then values are taken from the respective fields in the synthesizer GUI. Otherwise the latter are overwritten by the specified values.

<Path to audio file> can also be an URL of a remote audio source. Carrier frequencies must be specified in kHz.

Concrete example of a valid m3u-file:

```
#EXTM3U
#PLAYLIST: My favorites

#EXTPARAM_BW: 8
#EXTPARAM_RECLENGTH: 01:05:00

#EXTGRP: 783
#EXTINF: -1,Lud Gluskin Ambassadonians - Tiger Rag
C:\Users\user1\music\gr_000002567bk2.mp3
#EXTINF: -1,Comedian Harmonists – Greatest Hits -Vol.1
C:\Users\user1\music\Comedian Harmonists\04-Ich kuesse Ihre Hand, Madame.wav
#EXTINF: -1, Comedian Harmonists – Greatest Hits -Vol.1
C:\Users\user1\music\Comedian Harmonists\03-Tea For Two.wav

#PLAYLIST: My favorites

#EXTGRP: 900
#EXTINF: -1, Comedian Harmonists – Greatest Hits -Vol.1
C:\Users\user1\music\Comedian Harmonists\ul_000010387.mp3
#EXTINF: -1, Camerata Vocal Group - Just
```

C:\Users\user1\music\Camerata Vocal Group\Just\02-In a sentimental mood.mp3

#EXTINF: -1, Camerata Vocal Group - Just

C:\Users\user1\music\Camerata Vocal Group\Just\09-Oya Negra.mp3

#PLAYLIST: My favorite URLs

#EXTGRP: 1000

#EXTINF: -1,Lewitsch Tanzorchester - I've got a cross eyed papa

http://my.favorite.URL/Sound/title001.mp3

#EXTINF: -1,Lewitsch Tanzorchester - If you do - what you do

http://my.favorite.URL/Sound/title001/title002.mp3

While filling the playlist, several basic checks are performed and displayed in the status message window. This can take a while and may appear as cumbersome for the user. If such checks are deemed unnecessary, they can be obviated by checking the radio button 'suppress playtime' below the gain slider item (see 'advanced controls' in fig. 1-40).

If URLs or files are not accessible for some reasons, an error message will pop up and synthesis will be stopped

2 KNOWN BUGS

The following problems have been observed:

2.1 GENERAL:

If you try to open a file that is currently open in another application, the program crashes. Please always close recording or playback in all other applications first.

2.2 ANNOTATOR:

The SNR values are not always determined correctly. Some values are very low (1 – 5) though the SNR is strong (e.g. 50dB). Please check the values in the final yaml-files and correct, if necessary.

2.3 PLAYER:

Disconnecting the playback device may have unexpected consequences. Especially the fl2k mode leads to broken pipes and deadlocks . Then the COHIWizard either crashes directly or should be restarted anyway.

2.4 SYNTHESIZER

Instances of unexpected behavior have been observed in the synthesizer when changing the number of carriers in the middle of a project. Thus it is recommended to set the number of carriers at the beginning and leave it untouched, if possible.

3 ANNEX:

3.1 FILESYSTEM:

3.1.1 Installation file system

3.1.2 WINDOWS and LINUX: installation directory

From version 2.3 on, the installer and the setup tool create the following structure in the installation directory 'COHI_root' (default: %USERPROFILE%\COHIWizard under Windows, ~/COHIWizard under LINUX). Optional modules and drivers only appear if they have been selected in the setup tool:

```
COHI_root
├── COHIWizard.bat / COHIWizard.sh       start COHIWizard
├── COHIWizard_console.bat               (Windows) start with console window
├── COHIWizard_setup.bat / .sh           start the setup tool
├── python                               (Windows installer only) private Python
3.10
├── venv                                 virtual Python environment with all
packages
├── documentation
│   └── UserManual_COHIWizard.pdf
├── requirements.txt, LICENSE, README.md
└── sources
    ├── COHIWizard.py, cohi_setup.py, ...
    ├── config_wizard.yaml               settings (section 3.3)
    ├── config_modules.yaml              modules/drivers (section 3.2)
    ├── .cohi_setup_state.json           installation state of the setup tool
    ├── COHIWizard_console.log           messages when started without console
    ├── core, player, resampler, logos   mandatory modules
    ├── spectralviewer, wavheader_editor, yaml_editor,
    │   annotator, synthesizer           optional modules
    ├── dev_drivers
    │   ├── libs                         shared C++ DSP libraries (.so/.dll)
    │   ├── fl2k_universal, stemlab_universal, ADALM2000_universal
    │   └── fl2k_stream, stemlab_125_14, ADALM2000   (reference drivers)
    ├── ffmpeg-master-latest-win64-gpl-shared   (Windows) local ffmpeg
    ├── tools                            (Windows) Zadig
    └── out                              default recording path
```

This system must not be changed (e.g. by renaming), otherwise the software will not work correctly.

3.1.3 LINUX, Windows when using Python sources

If you use the sources by calling COHIWizard.py from the Python prompt, then for each module (Tab) the following additional directories / files are required:

```
/sources
      ├── module_#_name/
      │   ├── __init__.py
      │   ├── module_core.py
      │   ├── module_widget.py
      │   ├── <optional auxiliaries>
      │   │   ├── ressources
      │   │   │   ├── module_widget.ui
```

Typical modules are:

- resampler
- synthesizer
- wavheader_editor
- yaml_editor

New modules may be added in future versions.

3.1.4 Annotation files

A folder with the name of the recording and the prefix 'ANN_' is created in the recording directory (RECORDINGDIR). If the recording e.g. has the name 'Bisamberg_closure_1476_027.wav', a folder

ANN_Bisamberg_closure_1476_027.wav

appears.

The following files are created in this folder:

cohiradia_metadata.yaml
cohiradia_metadata_header.yaml
cohiradia_metadata_tailer.yaml
snrannotation.yaml
stations_list.yaml
status.yaml
COHI_YAML_FINAL.yaml

The current annotation status is saved in status.yaml, i.e. the information on where the program should continue in the event of interruptions. snrannotation.yaml and stations_list.yaml are temporary files required by the system. cohiradia_metadata_header.yaml and cohiradia_metadata_tailer.yaml contain the station data entered in the YAML editor, cohiradia_metadata.yaml is a temporary file and COHI_YAML_FINAL.yaml is the finished yaml file for the landing page.

3.2 CHANGING THE MODULES CONFIGURATION:

Normally you do not need to edit config_modules.yaml: the setup tool (section 1.2) writes it every time modules or drivers are installed or removed. Removing a tab with the setup tool is therefore the recommended way. A file written by the setup tool with all modules and the six drivers of v2.3 reads:

```
modules:
  spectralviewer: view_spectra
  resampler: resample
  wavheader_editor: wavheader_editor
  yaml_editor: yaml_editor
  annotator: annotate
  synthesizer: synthesizer
module_names:
  spectralviewer: View Spectra
  resampler: Resampler
  wavheader_editor: Wavheader Editor
  yaml_editor: Yaml Editor
  annotator: Annotator
  synthesizer: Synthesizer
dev_workers:
  fl2k_universal: cohi_playrecworker.py
  stemlab_universal: cohi_playrecworker.py
  ADALM2000_universal: cohi_playrecworker.py
  fl2k_stream: cohi_playrecworker.py
  stemlab_125_14: cohi_playrecworker.py
  ADALM2000: cohi_playrecworker.py
SDR_controllers:
  fl2k_universal: SDR_control.py
  (same driver names as under dev_workers)
```

If you nevertheless want to hide a tab manually, e.g. the yaml editor, the annotator and the synthesizer on a Raspberry Pi, remove the corresponding lines in the fields ‘modules’ and ‘module_names’, save the file and restart the COHIWizard. The resampler must not be removed, it is needed internally. Please note that the next run of the setup tool rewrites the file according to the modules installed at that time.

3.3 MANUAL SETTINGS IN THE CONFIG_WIZARD.YAML

The file config_wizard.yaml stores important default settings of the COHIWizard, such as the last used file paths, previously chosen settings like the IP-address of the STEMLAB and some parameters which are used during runtime. **Normally this file should not be touched**, because otherwise the COHIWIzard may not even start up any more. However, sometimes manual configurations are necessary. **Important:** before changing parameters, make a backup copy of the file so that you can restore it in case of failure.

3.3.1 Case 1: Adaptation of the parameters for the ADALM2000

Oversampling:
There is one parameter which determine the factor by which the Nyquist frequency is set:

relaxfactor_OSR: 1.2

The ADALM2000 is set to intrinsically sample with 75MS/s. For a bandlimit of, say, 1.8 MHz theoretically need at least 3.6 MS/s to fulfill the Nyquist criterion. This value should be set as low as reasonably possible, because it determines the processing load and the data transfer rate. The software automatically determines the lowest possible integer factor by which the 75MS/s could be divided so as still fulfill the Nyquist limit. This factor is called 'oversampling ratio' (OSR). In our case that would be 3.75MS/s, the factor then being 20. Thus the signal is effectively oversampled by 20. However, then the first alias already appears at 1.95 MHz, i.e. just 150 kHz above the end of the band. One would thus need an extremely steep filter to suppress the aliases. Therefore one can set a 'relaxation_factor' (relaxfactor_OSR) which reduces the OSR to the next allowed integer value. By default it is set to 1.2, which then yields ceil(20/1.2) = ceil(4.5) = 5 MS/s. The OSR is thus set to 15. Then the distance to the first alias is usually large enough to suppress the spurs with a filter of 4th order. The higher the factor, the less oversampling and the higher the computation load for the up-sampling procedure and the data transfer rate. Therefore we do not recommend values higher than 1.5. If the computer has poor performance, even at 1.2 there may occur signal interruptions. Setting

relaxfactor_OSR: 1

is then the lowest allowable value which could be tried, but then a very steep filter is required.

By the way: The OSR increases the effective bit resolution by sqrt(OSR). A value of 16 thus gives a factor of 4, which is 2 bits extra. Increasing the relaxation factor also reduces the oversampling and thus the effective bit resolution of the DAC.

AVC presetting

Sometimes the AVC may not work properly. There is another parameter 'volumefactor' which pre-scales the signal amplitude before sending to the ADALM2000. Usually a value of 1 is good. However, sometimes a reduction may be appropriate. The default setting is

volumefactor: 1

You can experiment with that setting, but usually the effect is limited.

3.3.2 Case 2: Adaptation of the skin

On first startup the COHIWizard is checking, if a display <= 7" is connected. If so, then it sets a different skin, as described in the Player section. If this behavior is not wanted, you can set

autoskin: false

By default autoskin is set true.

If you want to set another than the default skin, then change the index in the command:

skinindex: 5

skinindex can be set to 2, 3 or 5. 5 is the default PC skin, 2 and 3 are the skins for 7" displays. The former skins 0 and 1 were removed in v2.3; a config_wizard.yaml which still contains skinindex 0 or 1 starts with skin 5.

3.3.3 Case 3: Own recordings with digital AGC

If you want to record IQ files with the recording function (currently STEMLAB only), you may wish some AGC. This may reduce the effective bit resolution, as there is no hardware (controllable gain of an LNA). Anyway you may want to try it out. By default you find a line:

REC_AGC: false

Setting

REC_AGC: true

Will activate a digital AGC, but with rigidly coded parameters, so there is no setting of the AGC dynamics possible.

3.3.4 Case 4: Level and modulation of the audio carriers

The audio modulator of the universal drivers (section 1.5.2) uses the following entries. All of them are created with default values when config_wizard.yaml is generated for the first time:

```
audio_mix_level: 0.25    # base level of the audio carriers relative to the IQ
band
audio_mixer_db: 0        # position of the Mixer slider in dB (set by the GUI)
audio_mod_index: 0.8     # AM modulation index of the audio carriers
audio_rate_hz: 12500     # internal sampling rate of the audio sources (Hz)
```

The effective level of the audio carriers in the mode ‘IQ band and audio modulator’ is audio_mix_level × 10^(audio_mixer_db/20). For everyday use the Mixer slider is sufficient; audio_mix_level only needs to be changed if the useful range of the slider (±30 dB) does not fit. A higher audio_rate_hz improves the audio bandwidth of the carriers but increases the computing load, especially with many stations. The entries last_modulator_type, last_device_name and audioplaylist are written automatically and restore the last used modulator type, SDR device and station list at the next start.